



ARCH LINUX SERVER CONFIGURATION

- EcoGoods

Version 1

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1. Introduction

EcoGoods has acquired on-premises compute resources to deploy its website, product database and related services. This report details the CLI-only configuration of an Arch Linux virtual machine to fulfil the baremetal server specification, with no GUI installed. The system was configured to meet all requirements while applying least privilege, separation of duties and comprehensive verification.

2. Requirement 1: Virtual Machine Creation and Arch Linux Installation

2.1 Virtual Machine Specification

A virtual machine was created in VMware Workstation Pro (a Type-2 hypervisor) with 2 GB RAM, a 40 GB virtual disk split into multiple files, and NAT networking for package downloads (see Appendix A, Figures A.1-A.7). No GUI was installed; the system remains headless throughout.

The Arch ISO was booted into a live CLI environment, where connectivity, time synchronisation and disk detection confirmed package access and identified the virtual disk as `/dev/sda` (see Appendix B, Figures B.1-B.5).

2.2 Disk Partitioning and Formatting

The disk was partitioned using `cfdisk` with a DOS partition table, creating three partitions as specified (see Appendix C, Figures C.1-C.3):

Partition	Size	Mount Point	Filesystem
<code>/dev/sda1</code>	512 MB	<code>/boot</code>	ext4
<code>/dev/sda2</code>	14 GB	<code>/home</code>	ext4
<code>/dev/sda3</code>	~25.5 GB (remaining)	<code>/</code>	ext4

A DOS partition table was selected because the VM was configured for BIOS/Legacy boot mode. GPT was considered but requires an additional BIOS Boot Partition for GRUB embedding, making DOS simpler and more appropriate (Arch Linux, 2026a).

The ext4 filesystem was selected for all partitions due to its journalling, strong performance, and broad compatibility. While btrfs offers snapshots and compression and xfs performs well with large files, ext4 is the most stable and well-documented option for a general-purpose server (Ab Karim *et al.*, 2017). The partitions were formatted and mounted root first, then /home and /boot, so genfstab could detect all mount points. Final sizes and mounts were verified using lsblk and df -h (see Appendix C, Figures C.4-C.7).

2.3 Base System Installation

The pacman keyring was initialised and updated prior to installation to ensure package signature verification (see Appendix D, Figures D.1-D.2). The base system was installed via pacstrap with packages justified below (see Appendix D, Figure D.3):

Package	Justification
base, linux, linux-firmware, base-devel, networkmanager, nano, vi, curl	Core OS components, kernel, firmware, build tools, network management and CLI utilities (Arch Linux, 2026b).
sudo	Named-user privilege escalation, reducing direct root use.
openssh	SSH server daemon for secure remote access (OpenSSH, 2024).
grub	Bootloader required to boot from the installed disk (GNU GRUB, 2026).
acl	POSIX ACL management via setfacl and getfacl (Arch Linux, 2025).
docker, docker-compose	Container environment required by Requirement 7 (Docker, 2026a).

Nginx was excluded from the host system because it runs in Docker, eliminating a port 80 conflict. A persistent filesystem table was generated using `genfstab -U`, using UUID-based entries, ensuring correct mounting regardless of disk naming changes (see Appendix D, Figure D.4). The system was entered via `arch-chroot` to configure hostname, locale, timezone, and the GRUB (see Appendix D, Figures D.5-D.7). Post-reboot checks verified booting, mounts and services (see Appendix D, Figures D.8-D.12).

3. Requirement 2: Root Account Security

Root account security was implemented in stages to prevent accidental lockout. Direct root SSH access was disabled using `PermitRootLogin no` (see Appendix F, Figure F.4) and named administrator accounts handle privileged work via `sudo`. Once `sudo` access was confirmed, the root password was locked and verified (see Appendix F, Figure F.18).

Keeping root active with a strong password was rejected because a compromised credential grants immediate unrestricted access with no per-user logging, violating least privilege (Ma *et al.*, 2011).

4. Requirement 3 and 4: Administrative Users and Business User Accounts

4.1 Role-Based Group Design

Seven role-based groups were created before any user accounts, following least privilege (see Appendix E, Figure E.1). Assigning permissions to groups rather than individuals is industry best practice, keeping the system manageable as the organisation scales (Ma *et al.*, 2011):

Group	Purpose
sysadmins	Full administrative access
techleads	Elevated development access
srvadmin	Full access to all /srv directories
devteam	Full access to /srv/projects
webdev	Access to /srv/http only
testers	Read-only access to /srv/projects
sshusers	Controls SSH login

4.2 User Account Creation

Five user accounts were created and assigned to groups reflecting their roles within EcoGoods (see Appendix E, Figure E.1):

User	Account	Groups	Access Level
Akshita (Admin)	akshita	wheel, sysadmins, srvadmin, docker, sshusers	Full system administrator.
Ashley (IT Manager)	ashley	wheel, sysadmins, srvadmin, docker, sshusers	Full IT Manager admin.
Avery Wolfgang Stephenson	avery	techleads, srvadmin, devteam, webdev, docker, sshusers	Full /srv access and Docker.
Evelynn Hughes	evelynn	webdev, sshusers	SFTP to /srv/http only.
James Hunter	james	testers, sshusers	Read-only to /srv/projects.

Evelynn was assigned a standard bash shell rather than `/usr/sbin/nologin` because `nologin` blocks both interactive and SFTP sessions. Since she requires SFTP access to upload files to `/srv/http`, interactive CLI is instead blocked via the SSH `ForceCommand` directive, satisfying both requirements simultaneously (OpenSSH, 2024).

All accounts were verified with the `id` command (see Appendix E, Figure E.2). Administrative access for both `akshita` and `ashley` was configured by enabling the `wheel` group in `visudo` and `sudo` access was verified (see Appendix E, Figures E.3-E.4).

5. Requirement 5: SSH Server Configuration and Security

5.1 SSH Key Generation with PuTTYgen

SSH key authentication was configured using PuTTYgen with Ed25519 (see Appendix F, Figures F.1-F.2), which offers 3072-bit RSA-equivalent strength with smaller keys and faster computation (Siddiqua and Yousuf, 2026). The public key was deployed with restrictive permissions, while password authentication remained for non-key users, controlled by AllowGroups sshusers (see Appendix F, Figure F.3).

5.2 SSH Daemon Hardening

The SSH daemon was hardened with PermitRootLogin no, PubkeyAuthentication yes, PasswordAuthentication yes and AllowGroups sshusers (see Appendix F, Figure F.4). AllowGroups was chosen over individual AllowUsers entries because it scales more easily, as adding a user requires only group membership, not a configuration file change and service restart (OpenSSH, 2024).

A Match User block restricts Evelyn to SFTP-only access (see Appendix F, Figure F.5). The configuration was then validated using `sshd -t` before the service was restarted, preventing syntax errors from locking out all users (see Appendix F, Figure F.6).

5.3 SSH Access Testing and Verification

SSH access was verified for all accounts using PuTTY from the Windows host. The administrator account, akshita, authenticated successfully using Ed25519 key-based authentication and reached an interactive bash shell (see Appendix F, Figures F.7-F.8). Root login was refused, while ashley, avery and james authenticated using password authentication (see Appendix F, Figure F.9-F.12).

Evelyn's PuTTY connection was immediately dropped, confirming the Match User ForceCommand restriction (see Appendix F, Figure F.13), while WinSCP confirmed SFTP upload access to `/srv/http` with no terminal (see Appendix F, Figures F.14-F.17).

6. Requirement 6: File and Directory Permission Management

Three directories were created under /srv as follows:

Directory	Purpose	Primary group
/srv/http	Website files served by Nginx	webdev
/srv/projects/ecogoods	Development project source code	devteam
/srv/docker/ecogoods	Docker Compose and Nginx configuration	srvadmin

Standard chmod/chown permissions were insufficient as they only support the owner-group-others model and cannot assign different permission levels to multiple groups on the same directory (Arch Linux, 2025). ACL entries, default ACL entries and the setgid bit (2770) were therefore applied so newly created files automatically inherit correct group ownership and permissions (see Appendix G, Figure G.1). The final ownership, permissions and ACL rules were verified using ls -ld and getfacl (see Appendix G, Figures G.2- G.5).

Permission testing confirmed: Avery wrote to /srv/http and /srv/projects/ecogoods; Evelynn wrote to /srv/http but was denied the project directory; and James read project files but was denied write access (see Appendix G, Figures G.6- G.8).

7. Requirement 7: Docker Containerisation Environment

Docker was installed and enabled as a system service (see Appendix H, Figures H.1- H.3), with container lifecycle management restricted to sysadmins and techleads; denial was verified for other users (see Appendix H, Figure H.4). Docker Compose was used instead of individual docker run commands because Compose files are declarative, version-controllable and reproducible (Docker, 2026a). The following three containers were deployed and verified as running (see Appendix H, Figures H.7-H.10):

Container	Purpose	Exposure
ecogoods_mysql	Product database	Internal only, no host port
ecogoods_phpmyadmin	Database management interface	Via Nginx reverse proxy at /phpmyadmin
ecogoods_nginx	Website server	Host port 80

MySQL was not published to the host network. It used expose instead of ports on an internal Docker network, so only peer containers could access it, not external users (Docker, 2026b). This was verified using three independent methods (see Appendix H, Figure H.13).

Nginx was configured as the single external entry point on port 80 (see Appendix H, Figure H.15), routing / to the website and /phpmyadmin/ to phpMyAdmin. The /srv/http was mounted read-only into the container, enabling file serving while preventing host content modification (see Appendix H, Figures H.5- H.6).

Service testing confirmed that Nginx responded on port 80 and phpMyAdmin was accessible via the reverse proxy route, returning a valid HTTP response (see Appendix H, Figures H.11– H.12). Avery's database user avery_dbadmin successfully accessed the ecogoods database internally, confirming correct provisioning (see Appendix H, Figure H.14).

References

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- Siddiqua, M. and Yousuf, N. (2026) ‘A Comparative Analysis of RSA, ECDSA, and ED25519 in SSH Public Key Authentication’, *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox’s Bazar, Bangladesh, 19-21 December 2025. IEEE, pp. 3428–3433. Available at: <https://doi.org/10.1109/ICCIT68739.2025.11491667>

Appendix A - Virtual Machine Creation

WORKSTATION PRO™ 17

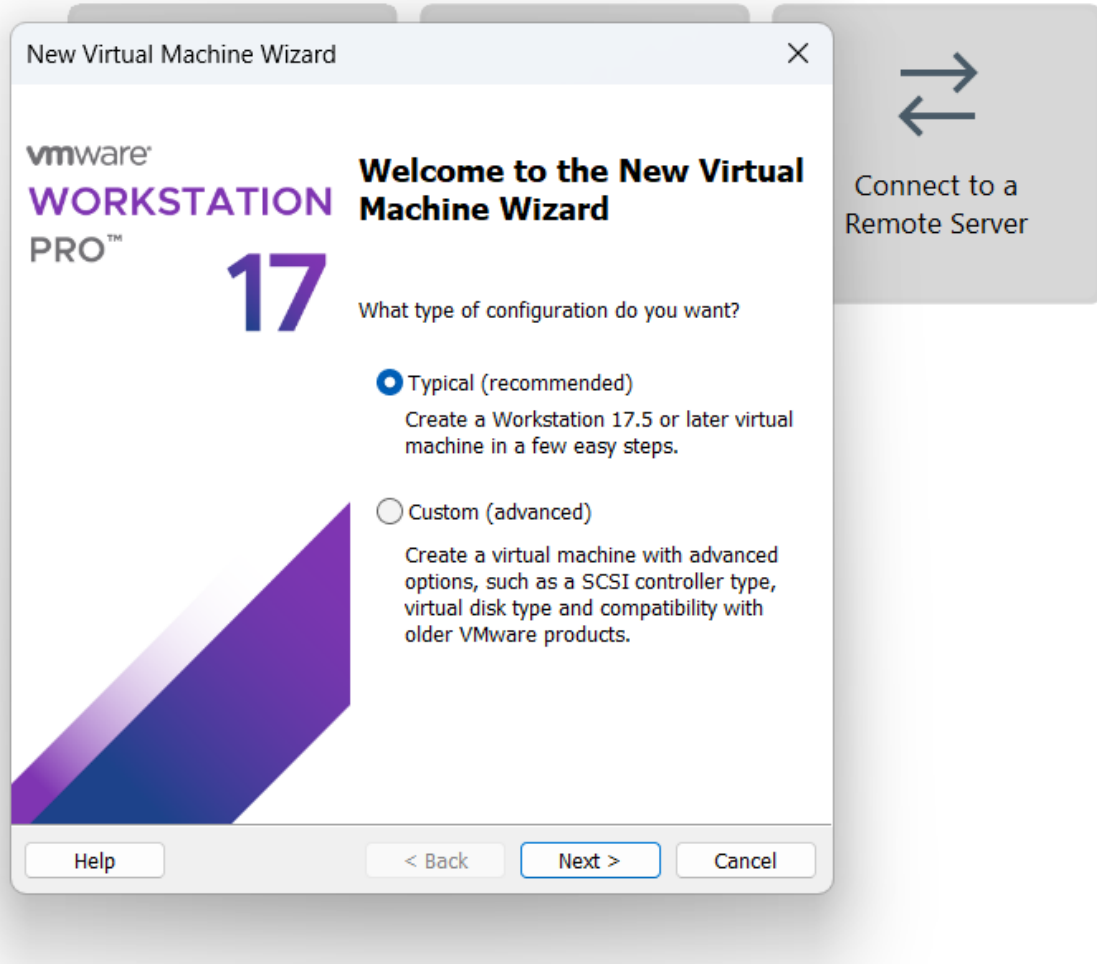


Figure A.1: New virtual machine wizard opened in VMware Workstation Pro

WORKSTATION PRO™ 17

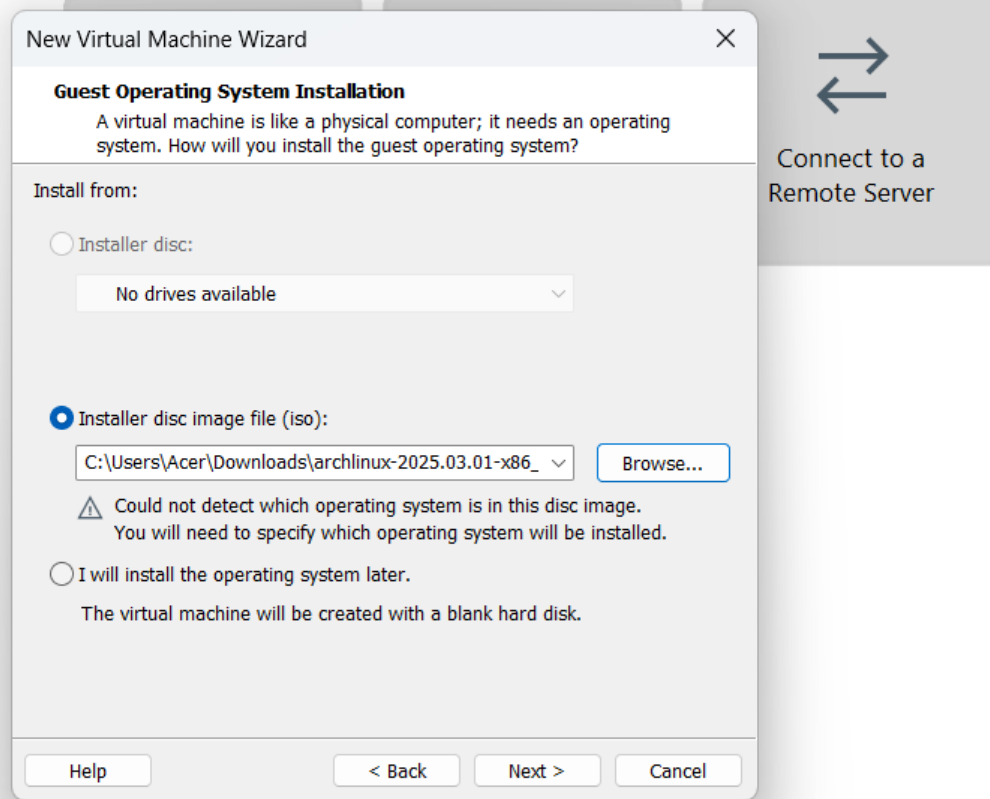


Figure A.2: Arch Linux 64-bit operating system selected

WORKSTATION PRO™ 17

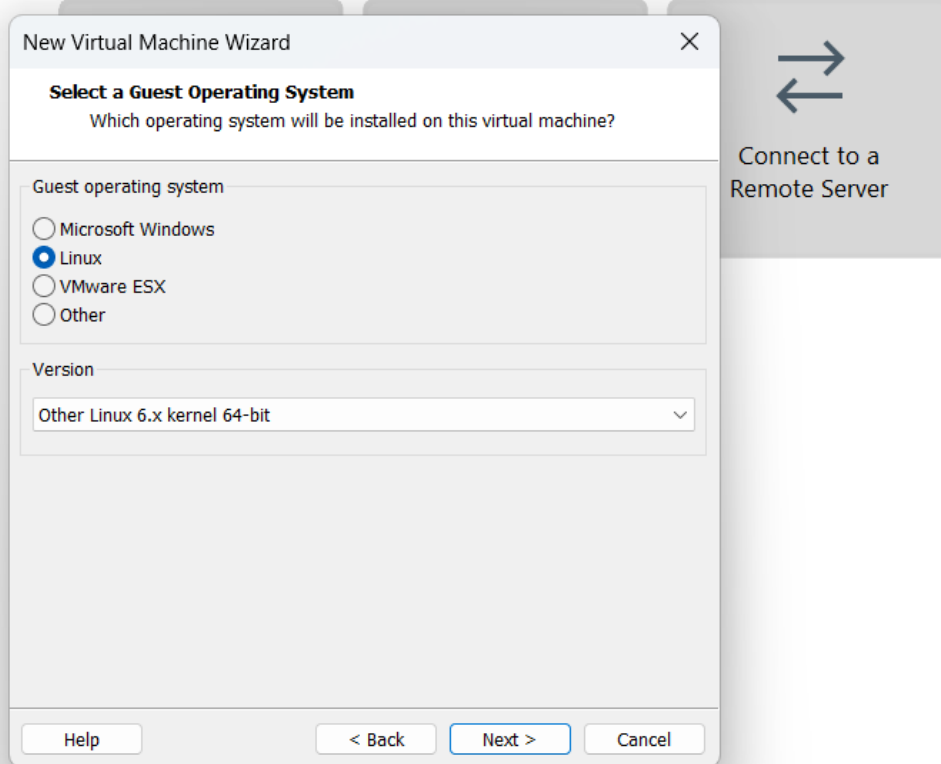


Figure A.3: Guest operating system selected

WORKSTATION PRO™ 17

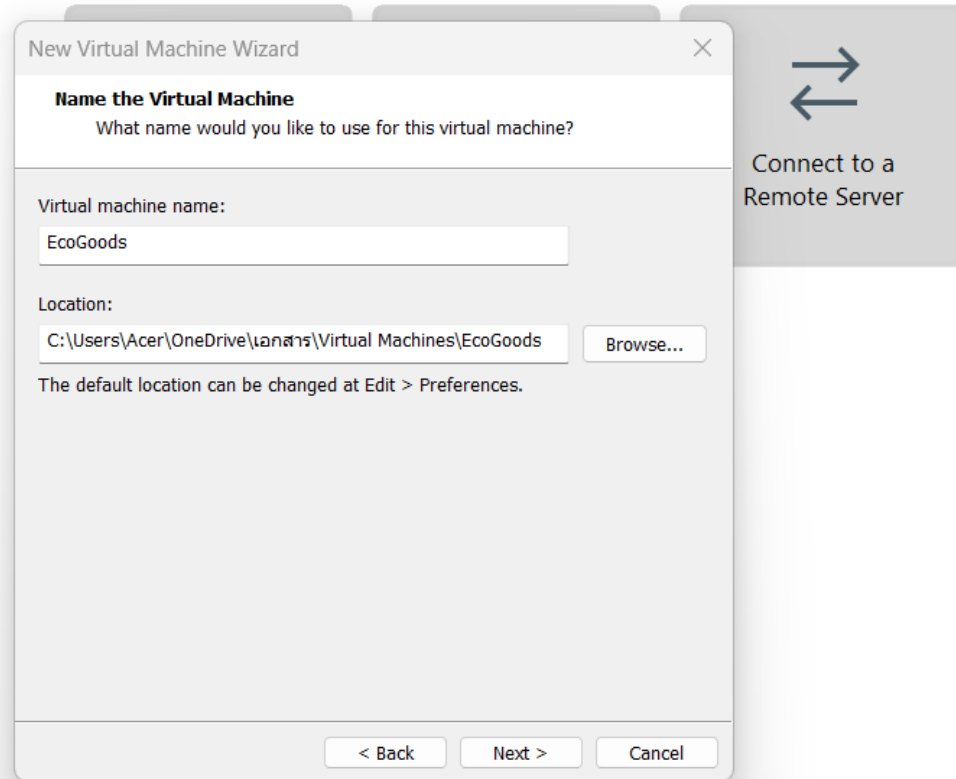


Figure A.4: Virtual machine name and storage location configured

WORKSTATION PRO™ 17

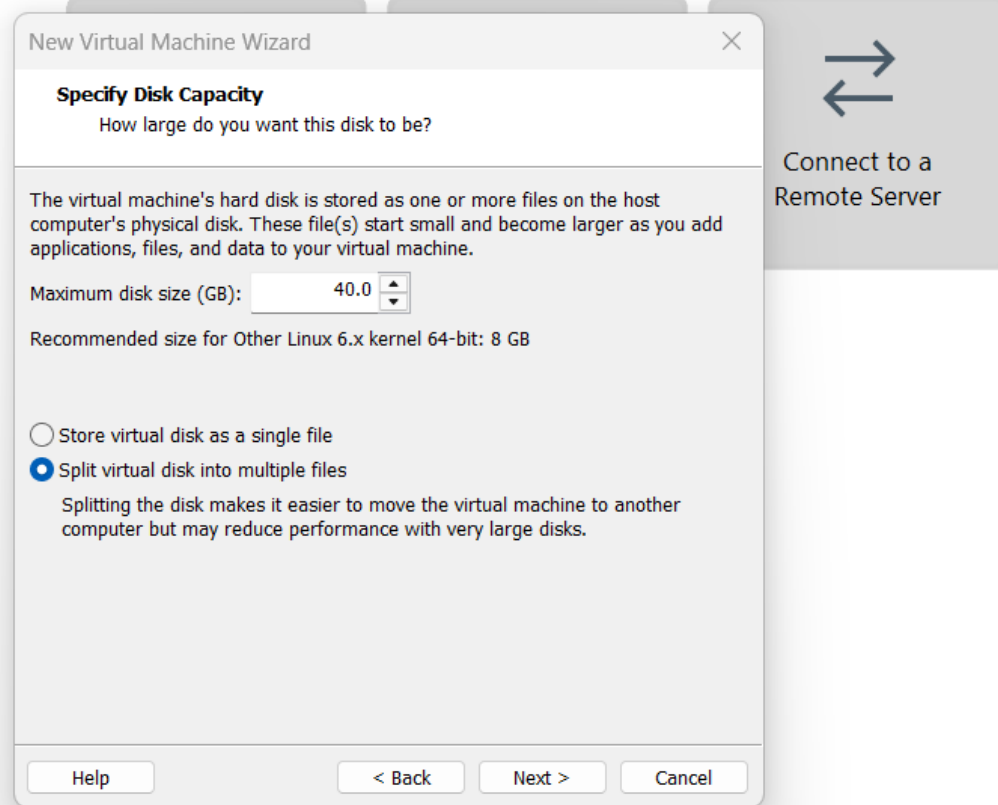


Figure A.5: Virtual disk configured as 40GB and disk split into multiple files

WORKSTATION PRO™ 17

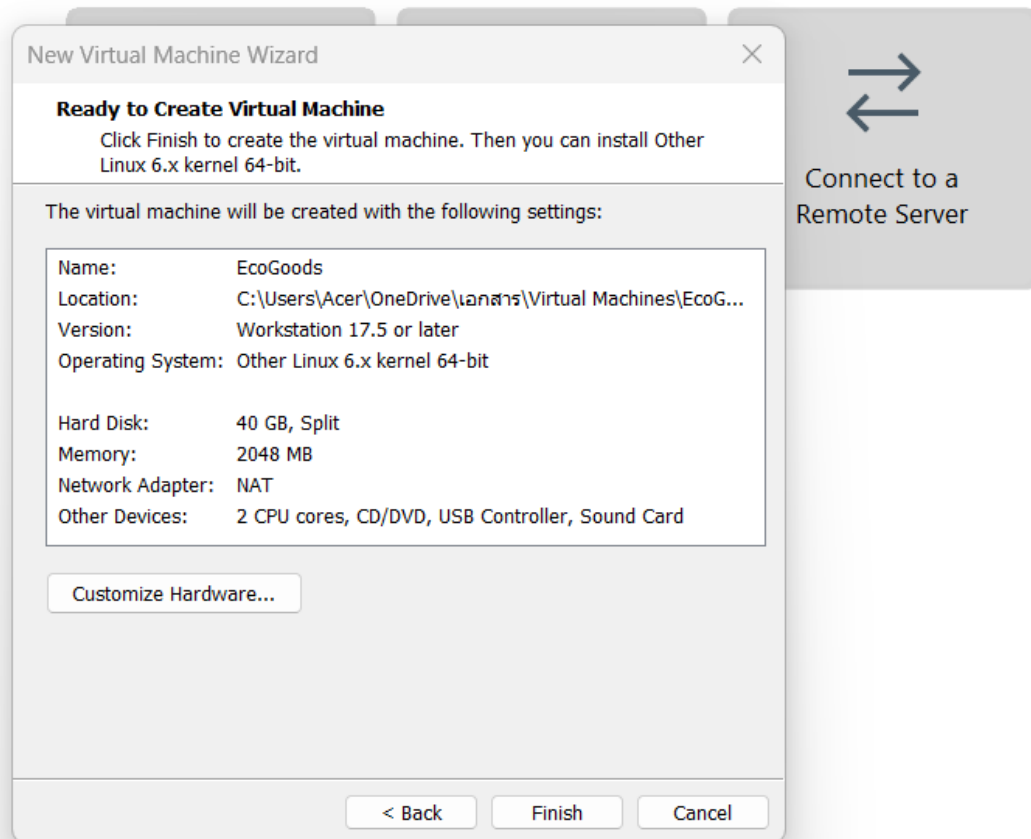


Figure A.6: Virtual machine specification overview



Figure A.7: Arch Linux BIOS boot menu

Appendix B - Live ISO Environment

```
Arch Linux 6.13.5-arch1-1 (tty1)

archiso login: root (automatic login)

To install Arch Linux follow the installation guide:
https://wiki.archlinux.org/title/Installation_guide

For Wi-Fi, authenticate to the wireless network using the iwctl utility.
For mobile broadband (WWAN) modems, connect with the mmcli utility.
Ethernet, WLAN and WWAN interfaces using DHCP should work automatically.

After connecting to the internet, the installation guide can be accessed
via the convenience script Installation_guide.

root@archiso ~ # ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:dd:df:ac brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29dddfac
    inet 192.168.25.140/24 metric 100 brd 192.168.25.255 scope global dynamic ens33
        valid_lft 1737sec preferred_lft 1737sec
    inet6 fe80::20c:29ff:fedd:dfac/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@archiso ~ # _
```

Figure B.1: Arch Linux live ISO successfully booted into CLI environment

```
valid_lft forever preferred_lft forever
root@archiso ~ # ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=32.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=34.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=37.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=41.1 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 32.767/36.653/41.134/3.146 ms
root@archiso ~ # _
```

Figure B.2: Verification of internet connectivity using ping 8.8.8.8

```

root@archiso ~ # ping -c 4 archlinux.org
PING archlinux.org (209.126.35.79) 56(84) bytes of data.
64 bytes from 209.126.35.79: icmp_seq=1 ttl=128 time=102 ms
64 bytes from 209.126.35.79: icmp_seq=2 ttl=128 time=107 ms
64 bytes from 209.126.35.79: icmp_seq=3 ttl=128 time=99.9 ms
64 bytes from 209.126.35.79: icmp_seq=4 ttl=128 time=103 ms

--- archlinux.org ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 8469ms
rtt min/avg/max/mdev = 99.852/102.809/107.041/2.634 ms
root@archiso ~ # █

```

Figure B.3: Verification of DNS/domain connectivity using ping archlinux.org

```

root@archiso ~ # timedatectl set-ntp true
root@archiso ~ # timedatectl status
          Local time: Sun 2026-05-10 06:33:15 UTC
          Universal time: Sun 2026-05-10 06:33:15 UTC
             RTC time: Sun 2026-05-10 06:33:16
            Time zone: UTC (UTC, +0000)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
root@archiso ~ # █

```

Figure B.4: Time synchronisation enabled using timedatectl set-ntp true and verified with timedatectl status

```
root@archiso ~ # lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0  7:0    0 836.7M  1 loop /run/archiso/airootfs
sda     8:0    0  40G   0 disk
sr0     11:0   1   1.2G   0 rom  /run/archiso/bootmnt
root@archiso ~ #
```

Figure B.5: Verification of virtual disk detection using lsblk

Appendix C - Disk Partitioning

```
l Select label type qqk
x gpt x
x dos x
x sgi x
x sun x
mqqqqqqqqqqqqqqqqqqqqqj
```

Figure C.1: DOS partition table selected in cfdisk

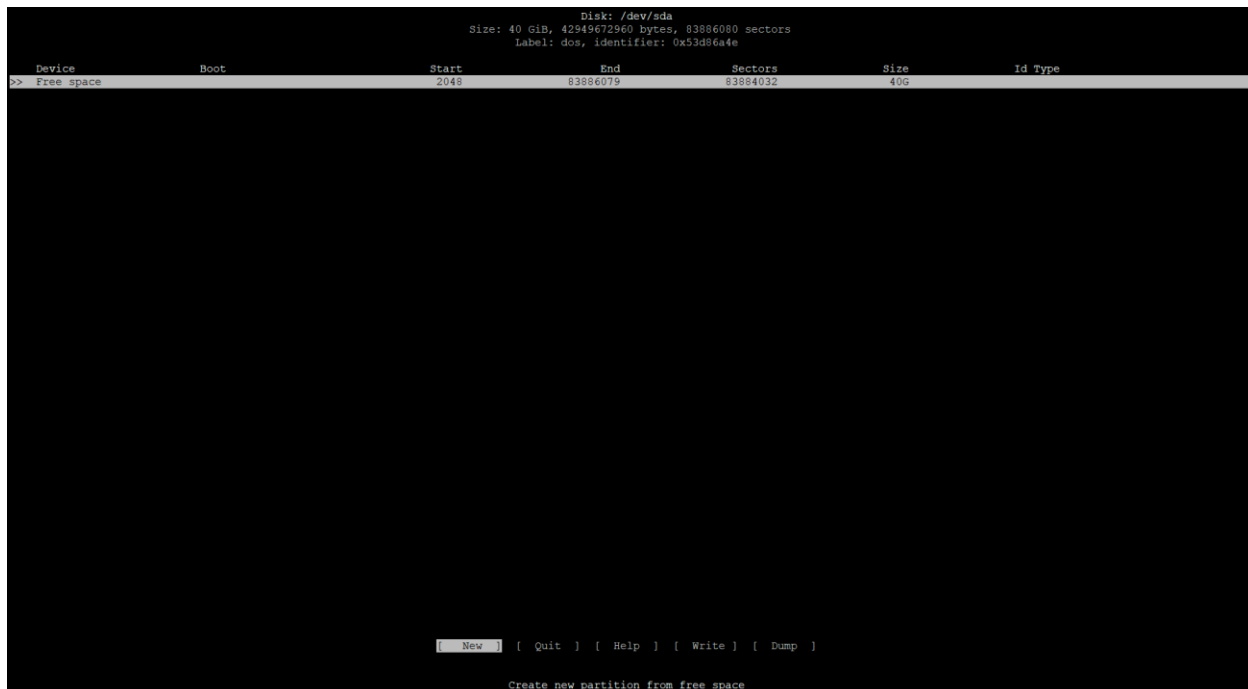


Figure C.2: cfdisk showing 40GB free space ready for partitioning

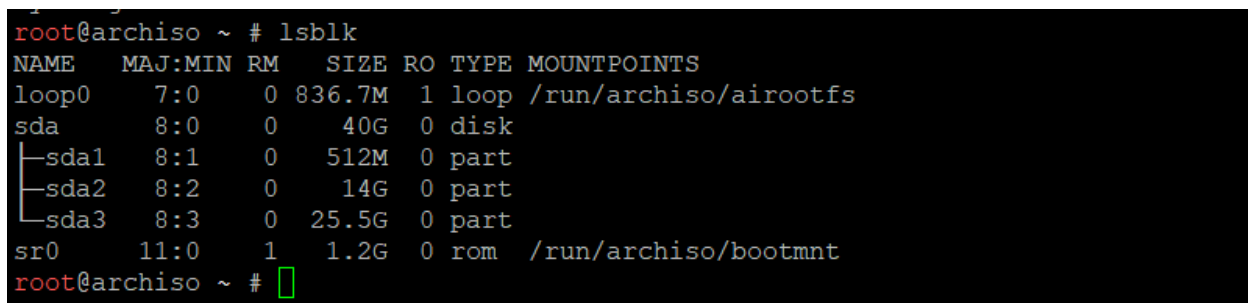


Figure C.3: Created partitions verified using lsblk

```

root@archiso ~ # mkfs.ext4 /dev/sda1
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 131072 4k blocks and 32768 inodes
Filesystem UUID: 523482bc-ec42-4214-9807-b22b88ec9e1d
Superblock backups stored on blocks:
    32768, 98304

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

root@archiso ~ # mkfs.ext4 /dev/sda2
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 3670016 4k blocks and 917504 inodes
Filesystem UUID: f44302ee-0afa-4151-aab4-a500efa47e86
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@archiso ~ # mkfs.ext4 /dev/sda3
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 6684416 4k blocks and 1671168 inodes
Filesystem UUID: cd252c0d-0acc-4d90-9612-bc147e340b4a
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done

root@archiso ~ # █

```

Figure C.4: Partitions formatted using mkfs.ext4

```

root@archiso ~ # mount /dev/sda3 /mnt
root@archiso ~ # mkdir /mnt/home
root@archiso ~ # mkdir /mnt/boot
root@archiso ~ # mount /dev/sda2 /mnt/home
root@archiso ~ # mount /dev/sda1 /mnt/boot

```

Figure C.5: Root, home and boot partitions mounted in the correct order

```

root@archiso ~ # lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0 836.7M  1 loop /run/archiso/airootfs
sda         8:0    0   40G   0 disk
├─sda1      8:1    0   512M   0 part /mnt/boot
├─sda2      8:2    0    14G   0 part /mnt/home
└─sda3      8:3    0  25.5G   0 part /mnt
sr0         11:0    1   1.2G   0 rom  /run/archiso/bootmnt
root@archiso ~ #

```

Figure C.6: Mount points verified using lsblk

```

root@archiso ~ # df -h
Filesystem      Size  Used Avail Use% Mounted on
dev             883M    0 883M   0% /dev
run            964M   9.2M 955M   1% /run
/dev/sr0        1.2G  1.2G    0 100% /run/archiso/bootmnt
cowspace        256M   1.1M 255M   1% /run/archiso/cowspace
/dev/loop0      837M  837M    0 100% /run/archiso/airootfs
airootfs        256M   1.1M 255M   1% /
tmpfs           964M    0 964M   0% /dev/shm
tmpfs           964M    0 964M   0% /tmp
tmpfs           1.0M    0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M    0 1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           1.0M    0 1.0M   0% /run/credentials/systemd-networkd.service
tmpfs           964M   2.7M 962M   1% /etc/pacman.d/gnupg
tmpfs           1.0M    0 1.0M   0% /run/credentials/getty@tty1.service
tmpfs           193M   8.0K 193M   1% /run/user/0
/dev/sda3        25G   2.1M  24G   1% /mnt
/dev/sda2        14G   2.1M  13G   1% /mnt/home
/dev/sda1        488M  152K 452M   1% /mnt/boot
root@archiso ~ #

```

Figure C.7: Disk partition and mount usage verified using df -h

Appendix D - Base System Installation, System Configuration and First Boot Verification

```
root@archiso ~ # pacman-key --init
root@archiso ~ # pacman-key --populate archlinux
==> Appending keys from archlinux.gpg...
==> Updating trust database...
gpg: next trustdb check due at 2026-06-08
root@archiso ~ # pacman -Sy archlinux-keyring
:: Synchronizing package databases...
core                               124.6 KiB   81.9 KiB/s   00:02 [#####] 100%
extra                             2.1 MiB    524 KiB/s   00:11 [#####] 25%
```

Figure D.1: Pacman keyring initialised and Arch Linux keys populated

```
root@archiso ~ # pacman -Sy archlinux-keyring
:: Synchronizing package databases...
core                               124.6 KiB   81.9 KiB/s   00:02 [#####] 100%
extra                             8.2 MiB    445 KiB/s   00:19 [#####] 100%

resolving dependencies...
looking for conflicting packages...

Packages (1) archlinux-keyring-20260420-1

Total Download Size:   1.21 MiB
Total Installed Size: 1.72 MiB
Net Upgrade Size:      0.05 MiB

:: Proceed with installation? [Y/n] y
:: Retrieving packages...
archlinux-keyring-20260420-1-any 1237.3 KiB   476 KiB/s   00:03 [#####] 100%
(1/1) checking keys in keyring [#####] 100%
(1/1) checking package integrity [#####] 100%
(1/1) loading package files [#####] 100%
(1/1) checking for file conflicts [#####] 100%
(1/1) checking available disk space [#####] 100%
:: Processing package changes...
(1/1) upgrading archlinux-keyring [#####] 100%
==> Appending keys from archlinux.gpg...
==> Locally signing trusted keys in keyring...
    -> Locally signed 1 key.
==> Importing owner trust values...
gpg: inserting ownertrust of 4
==> Disabling revoked keys in keyring...
    -> Disabled 10 keys.
==> Updating trust database...
gpg: Note: third-party key signatures using the SHA1 algorithm are rejected
gpg: (use option "--allow-weak-key-signatures" to override)
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid:   1  signed:   5  trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: depth: 1  valid:   5  signed:  86  trust: 0-, 0q, 0n, 5m, 0f, 0u
gpg: depth: 2  valid:  74  signed: 188  trust: 74-, 0q, 0n, 0m, 0f, 0u
gpg: next trustdb check due at 2026-10-21
:: Running post-transaction hooks...
(1/3) Reloading system manager configuration...
(2/3) Restarting marked services...
(3/3) Arming ConditionNeedsUpdate...
root@archiso ~ #
```

Figure D.2: Arch Linux keyring package updated using pacman -Sy archlinux-keyring

```

root@archiso ~ # pacstrap -K /mnt base base-devel linux linux-firmware nano vi sudo openssh networkmanager grub acl docker docker-compose curl
==> Creating install root at /mnt
==> Installing packages to /mnt
:: Synchronizing package databases...
core is up to date
extra is up to date
resolving dependencies...
:: There are 2 providers available for libxtables.so=12-64:
:: Repository core
1) iptables 2) iptables-legacy

Enter a number (default=1):
:: There are 3 providers available for initramfs:
:: Repository core
1) mkinitcpio
:: Repository extra
2) booster 3) dracut

Enter a number (default=1):
looking for conflicting packages...

Packages (207) archlinux-keyring-20260420-1 attr-2.5.2-2 audit-4.1.4-2 autoconf-2.73-1 automake-1.18.1-1 bash-5.3.9-1 binutils-2.46+r70+g155188eal0a7-1 bison-3.8.2-8 brotli-1.2.0-1
bzip2-1.0.8-6 ca-certificates-20240618-1 ca-certificates-mozilla-3.123.1-1 ca-certificates-utils-20240618-1 containerd-2.3.0-1 coreutils-9.11-1 cryptsetup-2.8.6-1
dbus-1.16.2-1 dbus-broker-37-3 dbus-broker-units-37-3 dbus-units-37-3 debugedit-5.2-1 device-mapper-2.03.40-1 diffutils-3.12-2 duktape-2.7.0-7
elfutils-0.189-1 expat-2.8.0-1 fakeroot-1.37.2-1 file-5.47-3 filesystem-2025.10.12-1 findutils-4.10.0-3 flex-2.6.4-6 gawk-5.4.0-1 gc-8.2.12-1
gcc-16.1.1+r24+g901eb00fa2c5-1 gcc-libs-16.1.1+r24+g901eb00fa2c5-1 gdm-1.26-2 gettext-1.0-2 glib2-2.80.1-1 glib3-2.43+r22+g9362e8ee10b2-2 gmp-6.3.0-3
gnulib-1.10n-20241231-1 gnupg-2.4.9-1 gnutls-3.8.13-1 gpvme-2.0.1-3 gpm-1.20.7.r38.ge82d1a6-6 grep-3.12-2 groff-1.24.1-1 guile-3.0.11-1 gzip-1.14-2 hwdmdata-0.407-1
==> Building image from preset: /etc/mkinitcpio.d/linux.preset: 'default'
==> Using default configuration file: '/etc/mkinitcpio.conf'
-> -k /boot/vmlinuz-linux -g /boot/initramfs-linux.img
==> Starting build: '7.0.5-arch1-1'
-> Running build hook: [base]
-> Running build hook: [systemd]
-> Running build hook: [autodetect]
-> Running build hook: [microcode]
-> Running build hook: [modconf]
-> Running build hook: [kms]
-> Running build hook: [keyboard]
-> Running build hook: [sgd-vconsole]
==> WARNING: sgd-vconsole: "/etc/vconsole.conf" not found, will use default values
-> Running build hook: [block]
-> Running build hook: [filesystems]
-> Running build hook: [fsck]
==> Generating module dependencies
==> Creating zstd-compressed initcpio image: '/boot/initramfs-linux.img'
-> Early uncompressed CPIO image generation successful
==> Initcpio image generation successful
(13/15) Reloading system bus configuration...
Skipped: Running in chroot.
(14/15) Checking for old perl modules...
(15/15) Updating the info directory file...
pacstrap -K /mnt base base-devel linux linux-firmware nano vi sudo openssh 19.56s user 19.02s system 10% cpu 6:21.38 total
root@archiso ~ #

```

Figure D. 3: Arch Linux base system and required packages installed using pacstrap

```

root@archiso ~ # genfstab -U /mnt >> /mnt/etc/fstab
root@archiso ~ # cat /mnt/etc/fstab
# Static information about the filesystems.
# See fstab(5) for details.

# <file system> <dir> <type> <options> <dump> <pass>
# /dev/sda3
UUID=cd252c0d-0acc-4d90-9612-bc147e340b4a / ext4 rw,relatime 0 1

# /dev/sda2
UUID=f44302ee-0afa-4151-aab4-a500efa47e86 /home ext4 rw,relatime 0 2

# /dev/sda1
UUID=523482bc-ec42-4214-9807-b22b88ec9e1d /boot ext4 rw,relatime 0 2

root@archiso ~ #

```

Figure D.4: Filesystem table generated using genfstab -U and verified with cat /mnt/etc/fstab

```

root@archiso ~ # arch-chroot /mnt
[root@archiso /]# ln -sf /usr/share/zoneinfo/Asia/Kathmandu /etc/localtime
[root@archiso /]# hwclock --systohc
[root@archiso /]# sed -i 's/^#en_GB.UTF-8 UTF-8/en_GB.UTF-8 UTF-8/' /etc/locale.gen
[root@archiso /]# locale-gen
Generating locales...
  en_GB.UTF-8... done
Generation complete.
[root@archiso /]# echo "LANG=en_GB.UTF-8" > /etc/locale.conf
[root@archiso /]# echo "ecogoods-server" > /etc/hostname
[root@archiso /]# cat > /etc/hosts <<'EOF'
> 127.0.0.1    localhost
::1          localhost
127.0.1.1    ecogoods-server.localdomain ecogoods-server
EOF
[root@archiso /]# █

```

Figure D.5: System configuration inside arch-chroot, including timezone, locale, hostname and /etc/hosts setup

```

[root@archiso /]# systemctl enable NetworkManager
Created symlink '/etc/systemd/system/multi-user.target.wants/NetworkManager.service' → '/usr/lib/systemd/system/NetworkManager.service'.
Created symlink '/etc/systemd/system/dbus-org.freedesktop.nm-dispatcher.service' → '/usr/lib/systemd/system/NetworkManager-dispatcher.service'.
Created symlink '/etc/systemd/system/network-online.target.wants/NetworkManager-wait-online.service' → '/usr/lib/systemd/system/NetworkManager-wait-online.service'.
[root@archiso /]# systemctl enable sshd
Created symlink '/etc/systemd/system/multi-user.target.wants/sshd.service' → '/usr/lib/systemd/system/sshd.service'.
[root@archiso /]# █

Created symlink /etc/systemd/system/multi-user.target.wants/sshd.service
[root@archiso /]# passwd
New password:
Retype new password:
passwd: password updated successfully
[root@archiso /]# █

```

Figure D.6: NetworkManager and SSH enabled, with the initial root password set

```

passwd: password updated successfully
[root@archiso /]# grub-install --target=i386-pc /dev/sda
Installing for i386-pc platform.
Installation finished. No error reported.
[root@archiso /]# grub-mkconfig -o /boot/grub/grub.cfg
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-linux
Found initrd image: /boot/initramfs-linux.img
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
[root@archiso /]# █

```

Figure D.7: GRUB installed for BIOS boot and boot configuration generated

```

Arch Linux 7.0.5-arch1-1 (tty1)

Try contacting this VM's SSH server via 'ssh usock%1709039532' from host.

ecogoods-server login:

```

Figure D.8: Successful first boot showing the ecogoods-server login prompt

```

[root@ecogoods-server ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   40G  0 disk
├─sda1       8:1    0   512M  0 part /boot
├─sda2       8:2    0    14G  0 part /home
└─sda3       8:3    0  25.5G  0 part /
sr0         11:0    1    1.2G  0 rom

[root@ecogoods-server ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda3        25G   2.8G   21G  12% /
devtmpfs         950M     0   950M   0% /dev
tmpfs            962M     0   962M   0% /dev/shm
tmpfs            385M  844K   384M   1% /run
none             1.0M     0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs            962M     0   962M   0% /tmp
/dev/sda1        488M   45M   407M  10% /boot
/dev/sda2        14G   2.1M   13G   1% /home
none             1.0M     0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs            193M   4.0K   193M   1% /run/user/0

[root@ecogoods-server ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:dd:df:ac brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29dddfac
    inet 192.168.25.140/24 brd 192.168.25.255 scope global dynamic noprefixroute ens33
        valid_lft 1023sec preferred_lft 1023sec
    inet6 fe80::b8c5:8eb6:544a:4d5f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@ecogoods-server ~]# █

```

Figure D.9: First boot verification using lsblk, df -h, and ip a

```

[root@ecogoods-server ~]# systemctl status NetworkManager --no-pager
● NetworkManager.service - Network Manager
   Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-10 13:30:25 +0545; 13min ago
     Invocation: 18978a93884b4c6eab87711129407e99
       Docs: man:NetworkManager(8)
    Main PID: 572 (NetworkManager)
      Tasks: 4 (limit: 2280)
     Memory: 38.2M (peak: 38.5M)
        CPU: 18%ms
    CGroup: /system.slice/NetworkManager.service
            └─572 /usr/bin/NetworkManager --no-daemon

May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.7979] dhcp4 (ens33): state changed new lease, address=192.168.25.140, acd pending
May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.9429] dhcp4 (ens33): state changed new lease, address=192.168.25.140
May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.9456] policy: set 'Wired connection 1' (ens33) as default for IPv4 routing and DNS
May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.9520] device (ens33): state change: ip-config -> ip-check (reason 'none', managed-type: 'full')
May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.9538] device (ens33): state change: ip-check -> secondaries (reason 'none', managed-type: 'full')
May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.9539] device (ens33): state change: secondaries -> activated (reason 'none', managed-type: 'full')
May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.9540] manager: NetworkManager state is now CONNECTED_SITE
May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.9541] device (ens33): Activation: successful, device activated.
May 10 13:30:26 ecogoods-server NetworkManager[572]: <info> [1778399126.9547] manager: startup complete
May 10 13:30:27 ecogoods-server NetworkManager[572]: <info> [1778399127.3858] manager: NetworkManager state is now CONNECTED_GLOBAL
[root@ecogoods-server ~]#

```

Figure D.10: NetworkManager service status verified as active

```

[root@ecogoods-server ~]# systemctl status sshd --no-pager
● sshd.service - OpenSSH Daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-10 13:41:00 +0545; 3min 38s ago
     Invocation: 6072dc50b7264732ba7db2f9276ff68b
       Docs: man:sshd(8)
            man:sshd_config(5)
    Main PID: 721 (sshd)
      Tasks: 1 (limit: 2280)
     Memory: 3.7M (peak: 21.5M)
        CPU: 51ms
    CGroup: /system.slice/sshd.service
            └─721 "sshd: /usr/bin/sshd -D [listener] 0 of 10-100 startups"

May 10 13:41:00 ecogoods-server systemd[1]: Starting OpenSSH Daemon...
May 10 13:41:00 ecogoods-server sshd[721]: Server listening on 0.0.0.0 port 22.
May 10 13:41:00 ecogoods-server sshd[721]: Server listening on :: port 22.
May 10 13:41:00 ecogoods-server systemd[1]: Started OpenSSH Daemon.
May 10 13:41:45 ecogoods-server sshd-session[736]: Accepted password for root from 192.168.25.1 port 56193 ssh2
May 10 13:41:45 ecogoods-server sshd-session[736]: pam_unix(sshd:session): session opened for user root(uid=0) by root(uid=0)
[root@ecogoods-server ~]#

```

Figure D.11: SSH service status verified as active

```
[root@ecogoods-server ~]# hostnamectl
  Static hostname: ecogoods-server
        Icon name: computer-vm
        Chassis: vm 
Chassis Asset Tag: No Asset Tag
        Machine ID: 14dad6aff23c47469881c8ed31e4e754
        Boot ID: 6edbd68609f247d3a3d55865f54535a1
        Product UUID: a2e54d56-d05b-4ce3-5322-4c3965dddfac
        AF_VSOCK CID: 1709039532
  Virtualization: vmware
  Operating System: Arch Linux
        Kernel: Linux 7.0.5-arch1-1
        Architecture: x86-64
  Hardware Vendor: VMware, Inc.
  Hardware Model: VMware Virtual Platform
  Hardware Serial: VMware-56 4d e5 a2 5b d0 e3 4c-53 22 4c 39 65 dd df ac
  Hardware Version: None
  Firmware Version: 6.00
        Firmware Date: Thu 2020-11-12
        Firmware Age: 5y 5month 3w 5d
[root@ecogoods-server ~]#
```

Figure D.12: Hostname and system information verified using hostnamectl

Appendix E - User and Group Administration

```
[root@ecogoods-server ~]# groupadd sysadmins
[root@ecogoods-server ~]# groupadd techleads
[root@ecogoods-server ~]# groupadd srvadmin
[root@ecogoods-server ~]# groupadd devteam
[root@ecogoods-server ~]# groupadd webdev
[root@ecogoods-server ~]# groupadd testers
[root@ecogoods-server ~]# groupadd sshusers
[root@ecogoods-server ~]# useradd -m -G wheel,sysadmins,srvadmin,docker,sshusers -s /bin/bash akshita
[root@ecogoods-server ~]# useradd -m -G wheel,sysadmins,srvadmin,docker,sshusers -s /bin/bash ashley
[root@ecogoods-server ~]# useradd -m -G techleads,srvadmin,devteam,webdev,docker,sshusers -s /bin/bash avery
password: password updated successfully
[root@ecogoods-server ~]# useradd -m -G webdev,sshusers -s /bin/bash evelynn
[root@ecogoods-server ~]# useradd -m -G testers,sshusers -s /bin/bash james
```

Figure E.1: Role-based groups and user accounts created

```
[root@ecogoods-server ~]# id akshita
uid=1000(akshita) gid=1007(akshita) groups=1007(akshita),998(wheel),970(docker),1000(sysadmins),1002(srvadmin),1006(sshusers)
[root@ecogoods-server ~]# id ashley
uid=1001(ashley) gid=1008(ashley) groups=1008(ashley),998(wheel),970(docker),1000(sysadmins),1002(srvadmin),1006(sshusers)
[root@ecogoods-server ~]# id avery
uid=1002(avery) gid=1009(avery) groups=1009(avery),970(docker),1001(techleads),1002(srvadmin),1003(devteam),1004(webdev),1006(sshusers)
[root@ecogoods-server ~]# id evelynn
uid=1003(evelynn) gid=1010(evelynn) groups=1010(evelynn),1004(webdev),1006(sshusers)
[root@ecogoods-server ~]# id james
uid=1004(james) gid=1011(james) groups=1011(james),1005(testers),1006(sshusers)
[root@ecogoods-server ~]#
```

Figure E.2: User and group membership verified using id

```
##
## User privilege specification
##
root ALL=(ALL:ALL) ALL

## Uncomment to allow members of group wheel to execute any command
%wheel ALL=(ALL:ALL) ALL

## Same thing without a password
# %wheel ALL=(ALL:ALL) NOPASSWD: ALL

## Uncomment to allow members of group sudo to execute any command
# %sudo ALL=(ALL:ALL) ALL

## Uncomment to allow any user to run sudo if they know the password
## of the user they are running the command as (root by default).
```

Figure E.3: wheel group enabled in visudo for sudo access

```
[root@ecogoods-server ~]# su - akshita
[akshita@ecogoods-server ~]$ sudo whoami

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for akshita:
root
[akshita@ecogoods-server ~]$
```

Figure E.4: Administrative sudo access verified using sudo whoami

Appendix F - SSH Hardening, PuTTY Key Authentication, SFTP Restriction and Root Account Locking

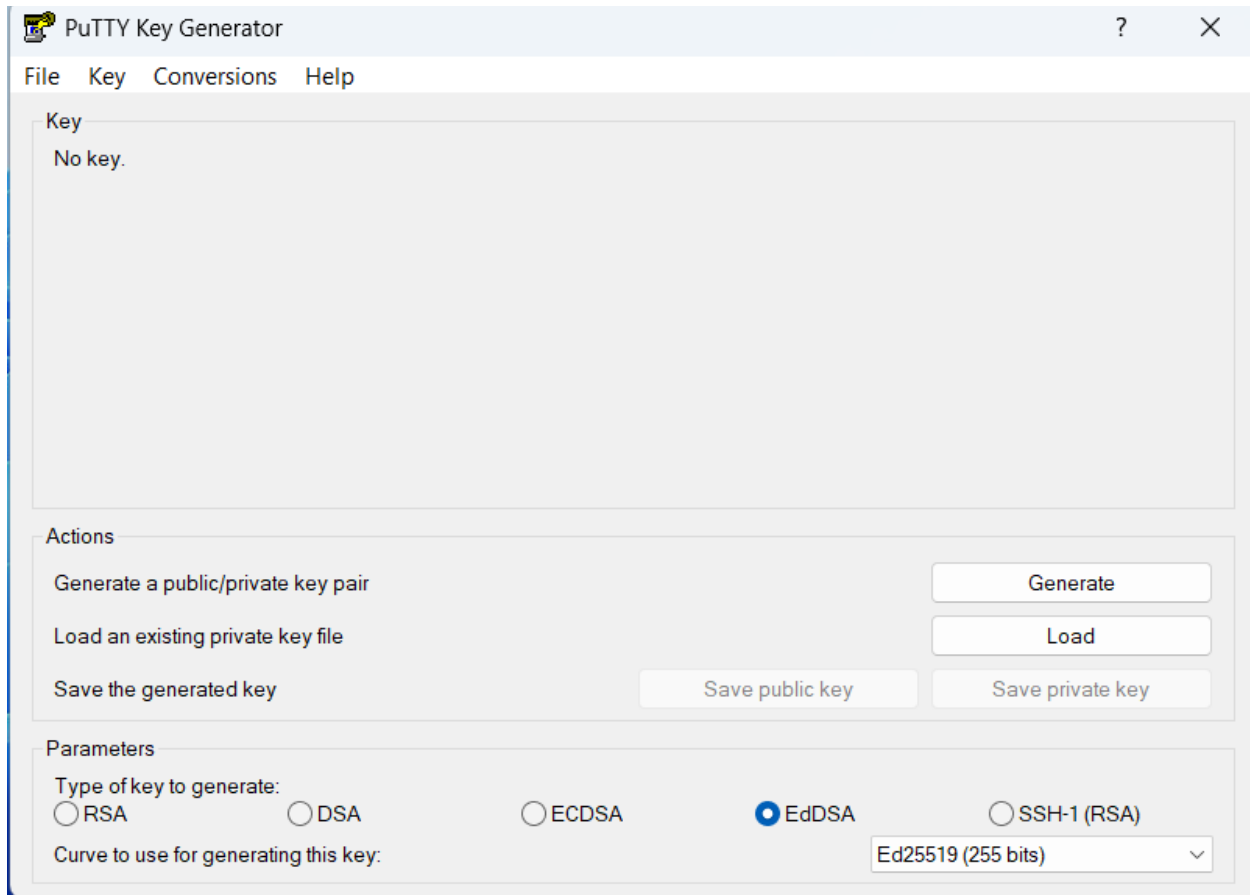


Figure F.1: EdDSA key type selected in PuTTYgen

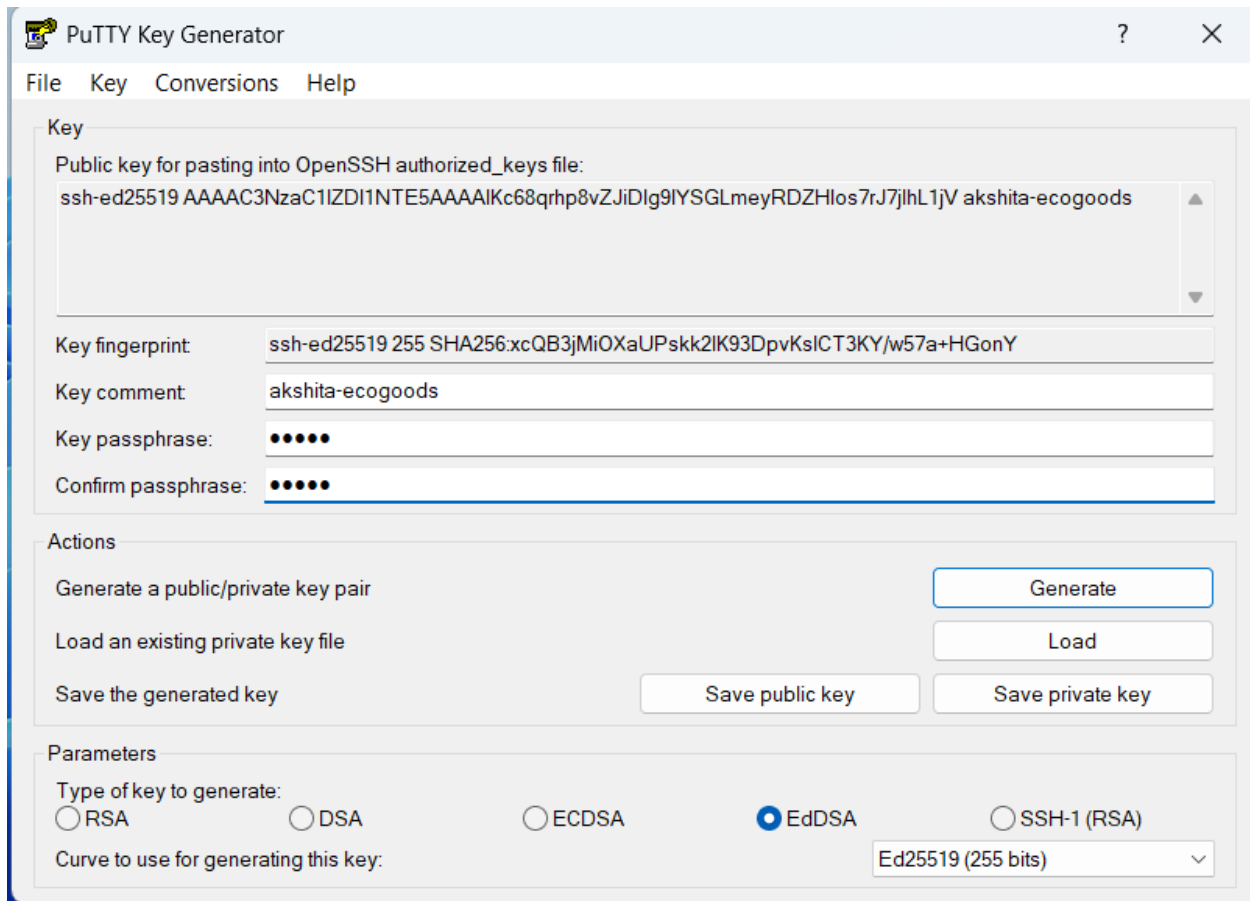


Figure F.2: SSH key pair generated in PuTTYgen with key passphrase configured

```
[root@ecogoods-server ~]# mkdir -p /home/akshita/.ssh
```

```
[root@ecogoods-server ~]# chmod 700 /home/akshita/.ssh
```

```
[root@ecogoods-server ~]# nano /home/akshita/.ssh/authorized_keys
```

```
[root@ecogoods-server ~]#
```

```
GNU nano 9.0 /home/akshita/.ssh/authorized_keys Modified  
<11ZDI1NTE5AAAAIKc68qrhp8vZJiD1g9lYSGLmeyRDZHlos7rJ7j1hL1jV akshita-ecogoods
```

```
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
[root@ecogoods-server ~]# chmod 600 /home/akshita/.ssh/authorized_keys  
chown -R akshita:akshita /home/akshita/.ssh
```

Figure F.3: Administrator SSH key added to `authorized_keys` with restrictive permissions

```

GNU nano 9.0 /etc/ssh/sshd_config
LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes
AllowGroups sshusers

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to "no" here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to "no" to disable keyboard-interactive authentication. Depending on
# the system's configuration, this may involve passwords, challenge-response,
# one-time passwords or some combination of these and other methods.
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes

^G Help      ^O Write Out  ^W Where Is   ^X Cut        ^E Execute    ^C Location   ^U Undo       ^M Set Mark   ^_ To Bracket  ^P Previous   ^_ Back
^X Exit      ^R Read File  ^A Replace    ^V Paste      ^J Justify    ^G Go To Line ^R Redo       ^N Copy       ^S Where Was  ^N Next       ^F Forward

```

Figure F.4: SSH configuration updated to disable root login and allow authorised SSH users

```

# Example of overriding settings on a per-user basis
#Match User anoncvs
#
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
Match User evelynn
    ForceCommand internal-sftp
    PermitTTY no
    X11Forwarding no
    AllowTcpForwarding no

```

Figure F.5: Match User evelynn block added to restrict Evelynn to SFTP-only access

```
[root@ecogoods-server ~]# sshd -t
[root@ecogoods-server ~]# systemctl restart sshd
[root@ecogoods-server ~]# systemctl status sshd --no-pager
● sshd.service - OpenSSH Daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-10 15:30:33 +0545; 18s ago
 Invocation: 45363243a4ea4bfeb2782f73fbd3aea2
    Docs: man:sshd(8)
          man:sshd_config(5)
 Main PID: 1746 (sshd)
   Tasks: 1 (limit: 2280)
  Memory: 1.2M (peak: 2M)
     CPU: 7ms
    CGroup: /system.slice/sshd.service
            └─1746 "sshd: /usr/bin/sshd -D [listener] 0 of 10-100 startups"

May 10 15:30:33 ecogoods-server systemd[1]: Starting OpenSSH Daemon...
May 10 15:30:33 ecogoods-server sshd[1746]: Server listening on 0.0.0.0 port 22.
May 10 15:30:33 ecogoods-server sshd[1746]: Server listening on :: port 22.
May 10 15:30:33 ecogoods-server systemd[1]: Started OpenSSH Daemon.
[root@ecogoods-server ~]#
```

Figure F.6: SSH configuration validated and sshd service restarted successfully

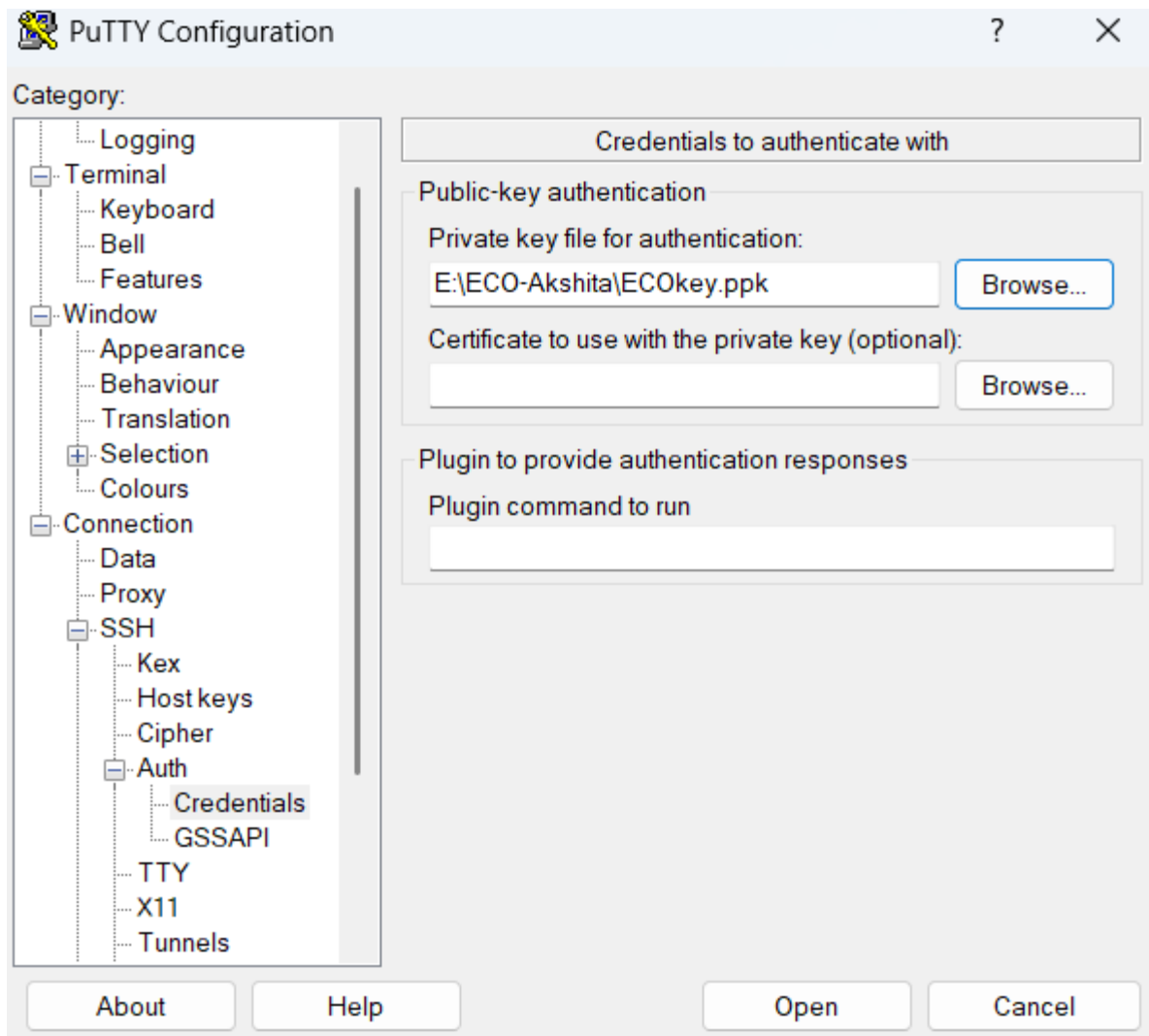
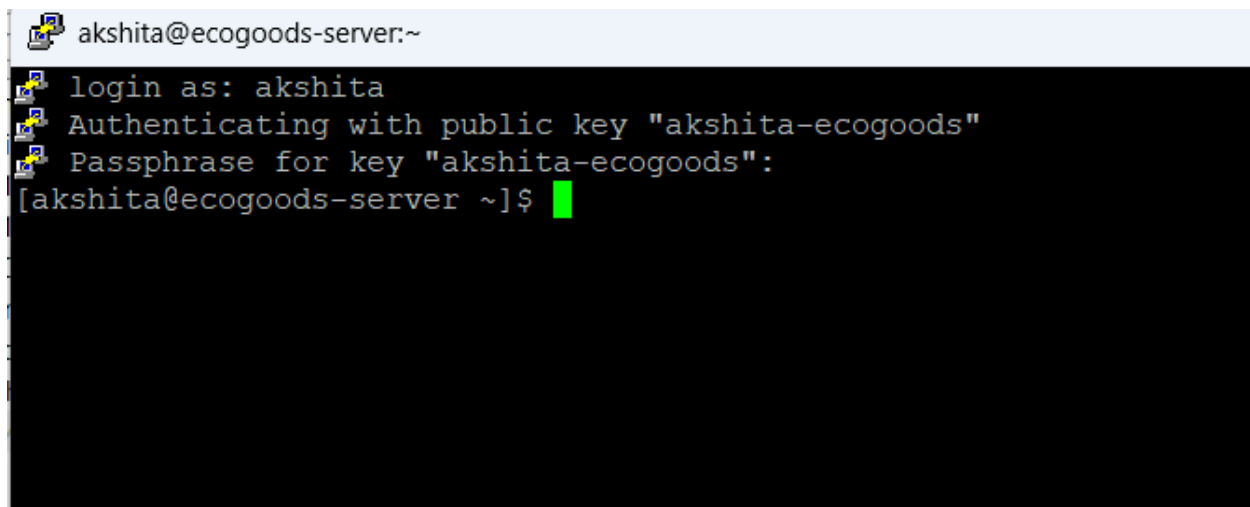
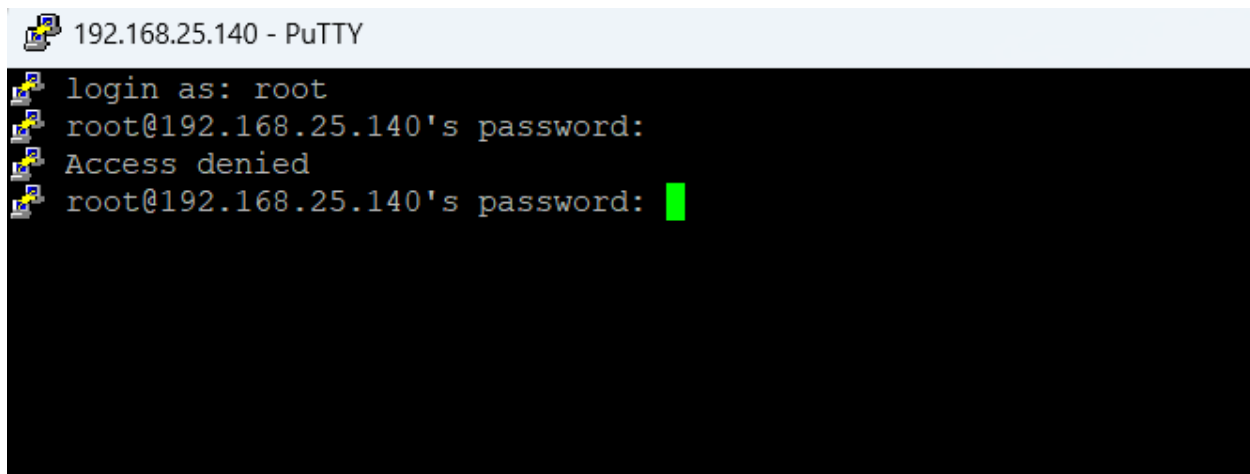


Figure F.7: PuTTY configured to use the administrator's private key



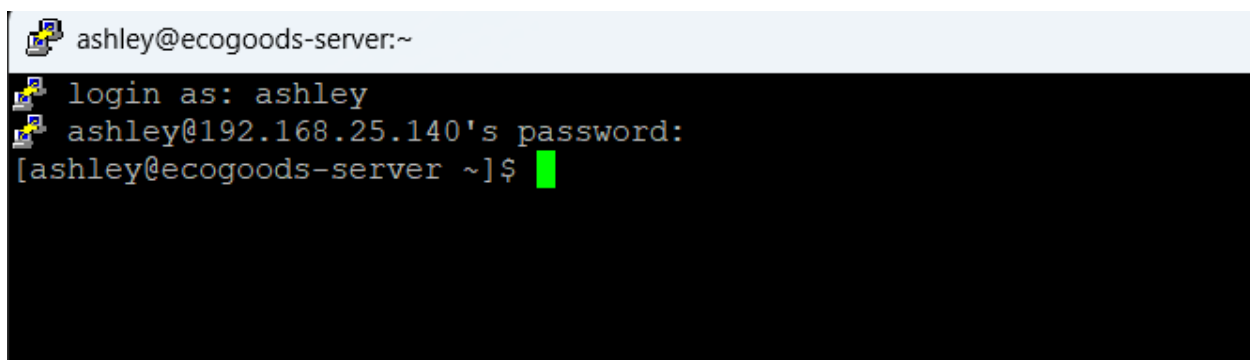
```
akshita@ecogoods-server:~  
login as: akshita  
Authenticating with public key "akshita-ecogoods"  
Passphrase for key "akshita-ecogoods":  
[akshita@ecogoods-server ~]$
```

Figure F.8: Successful PuTTY key-based login as the administrator account



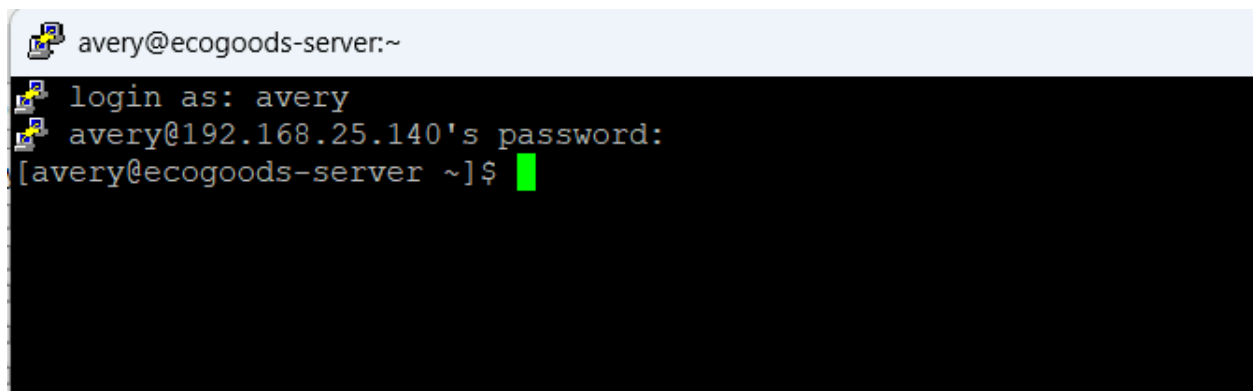
```
192.168.25.140 - PuTTY  
login as: root  
root@192.168.25.140's password:  
Access denied  
root@192.168.25.140's password:
```

Figure F.9: Root SSH login denied in PuTTY



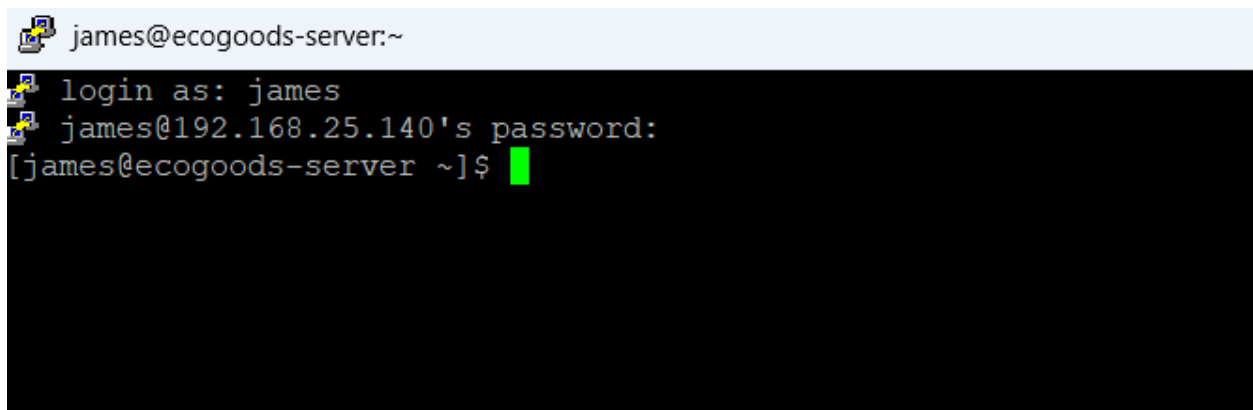
```
ashley@ecogoods-server:~  
login as: ashley  
ashley@192.168.25.140's password:  
[ashley@ecogoods-server ~]$
```

Figure F.10: Successful PuTTY login as Ashley



```
avery@ecogoods-server:~  
login as: avery  
avery@192.168.25.140's password:  
[avery@ecogoods-server ~]$
```

Figure F.11: Successful PuTTY login as Avery



```
james@ecogoods-server:~  
login as: james  
james@192.168.25.140's password:  
[james@ecogoods-server ~]$
```

Figure F.12: Successful PuTTY login as James

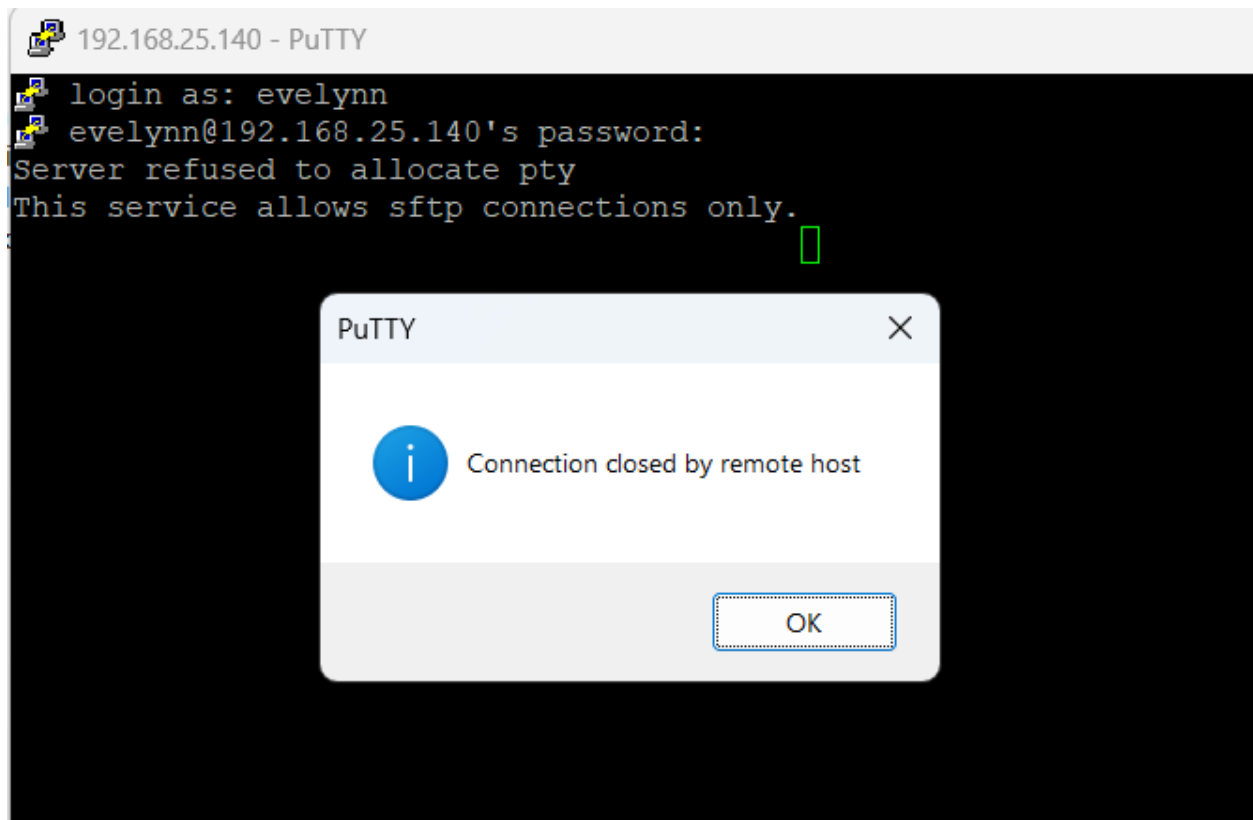


Figure F.13: Evelynn denied interactive shell access in PuTTY

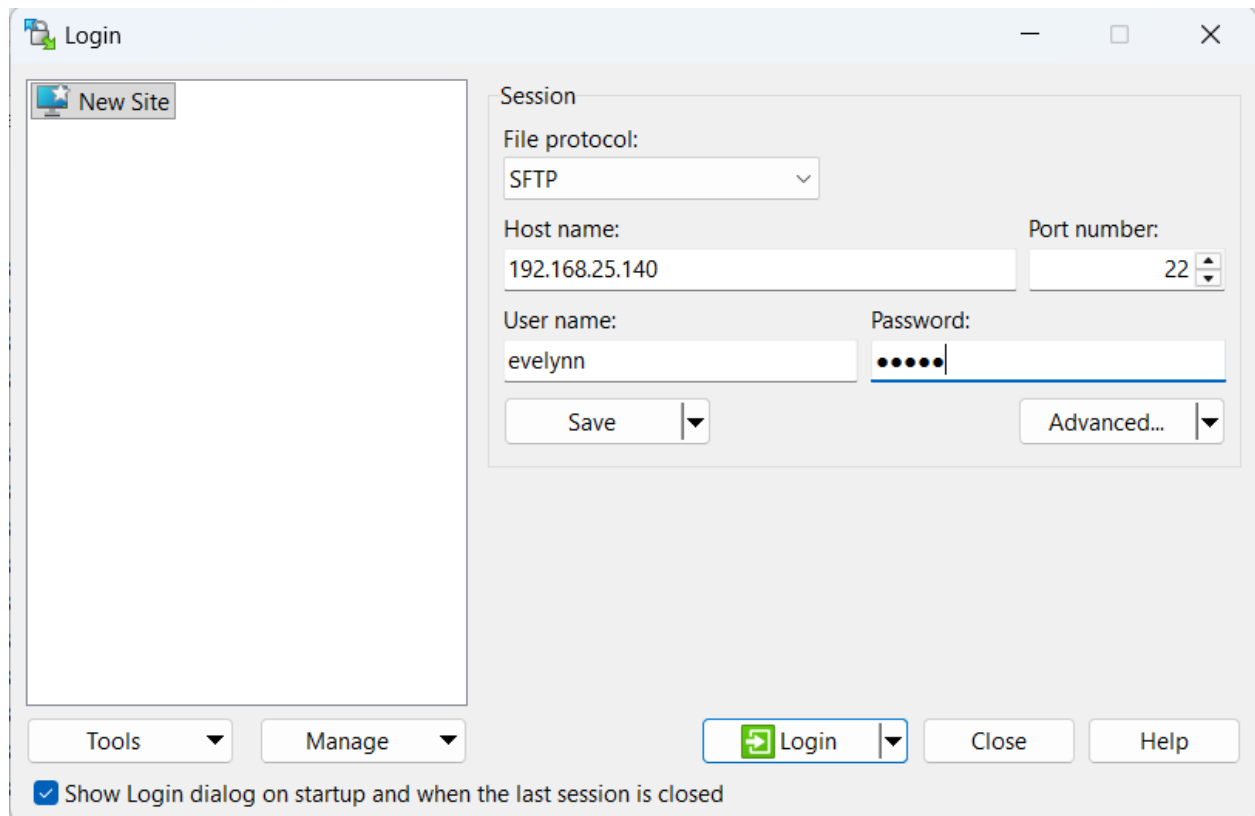


Figure F.14: Evelynn SFTP login initiated successfully in WinSCP

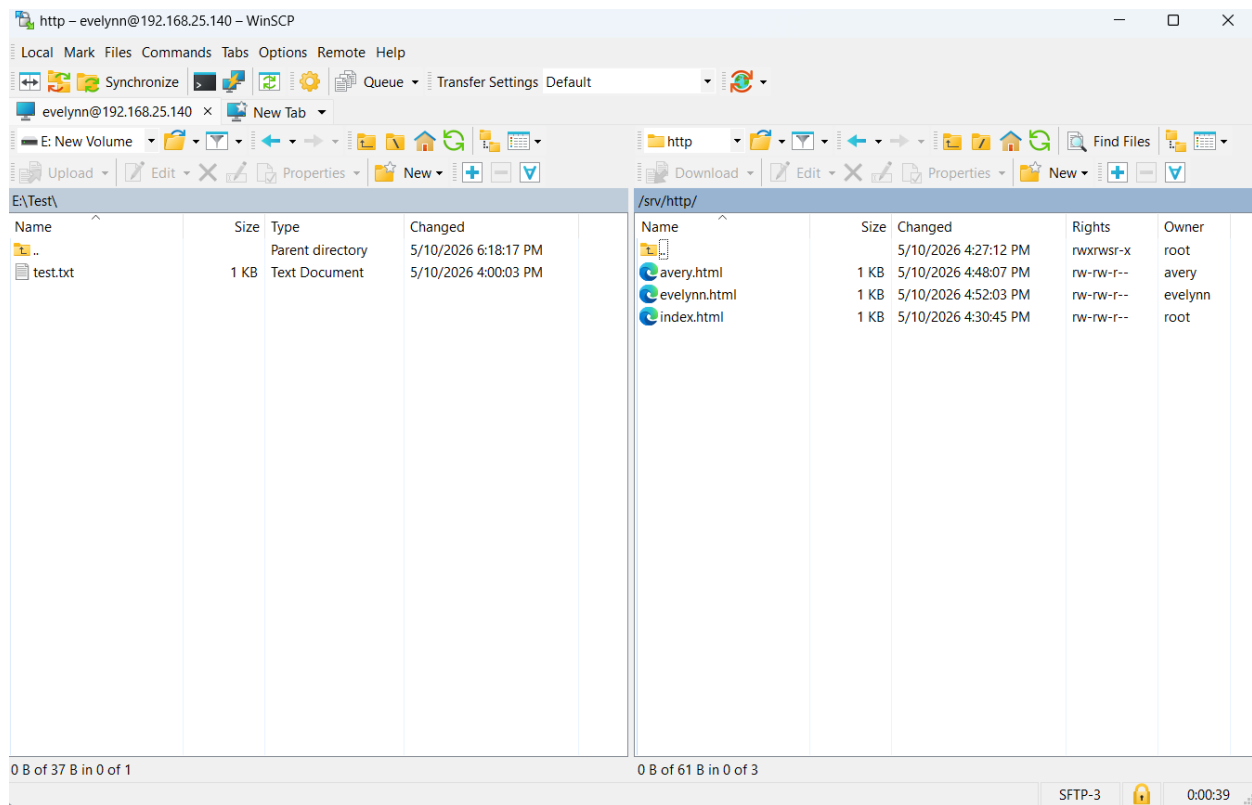


Figure F.15: WinSCP showing Evelynn’s access to /srv/http website files

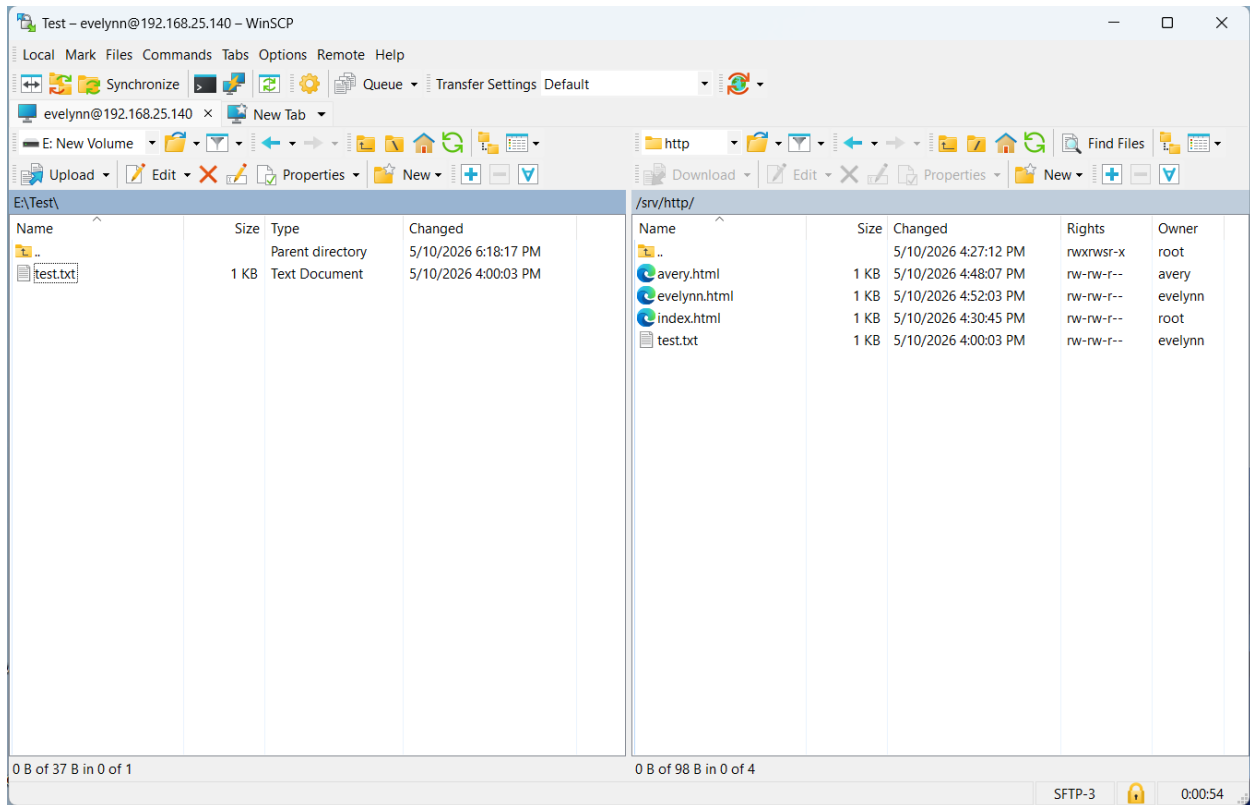


Figure F.16: Test file successfully uploaded to /srv/http using Evelynn's SFTP access

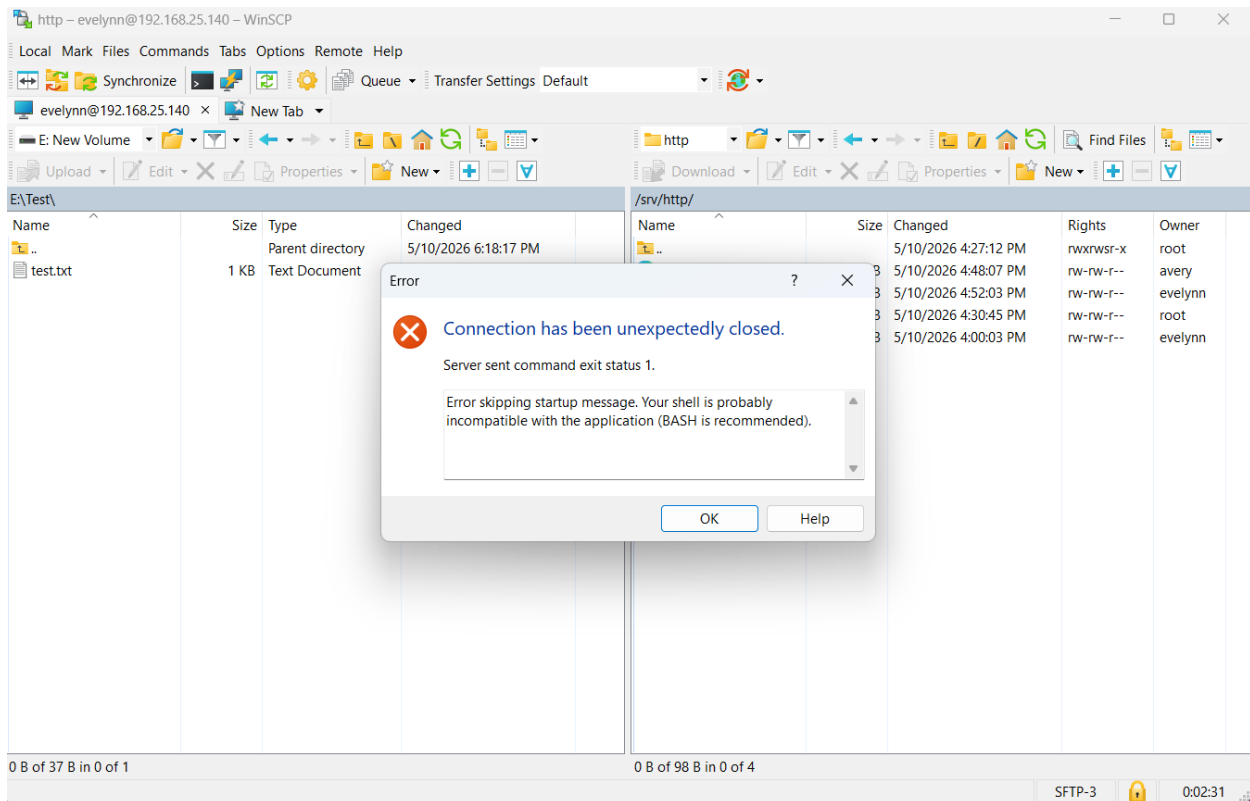


Figure F.17: WinSCP terminal access denied for Evelynn

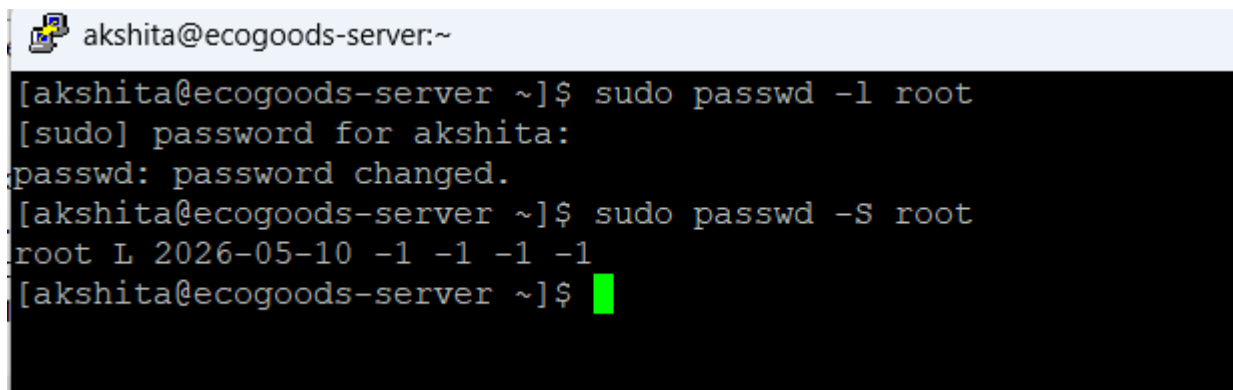


Figure F.18: Root account locked using passwd -l root and verified with passwd -S root

```
[akshita@ecogoods-server ~]$ sudo passwd -S root
root L 2026-05-10 -1 -1 -1 -1
[akshita@ecogoods-server ~]$ grep -E 'PermitRootLogin|PasswordAuthentication|AllowGroups' /etc/ssh/sshd_config
PermitRootLogin no
AllowGroups sshusers
PasswordAuthentication yes
# PasswordAuthentication. Depending on your PAM configuration,
# the setting of "PermitRootLogin prohibit-password".
# PAM authentication, then enable this but set PasswordAuthentication
[akshita@ecogoods-server ~]$
```

Figure F.19: Final SSH hardening verification showing root locked and secure SSH settings

Appendix G - ACL Permission Configuration and Permission Testing

```
[akshita@ecogoods-server ~]$ sudo mkdir -p /srv/http
[sudo] password for akshita:
[akshita@ecogoods-server ~]$ sudo mkdir -p /srv/projects/ecogoods
[akshita@ecogoods-server ~]$ sudo mkdir -p /srv/docker/ecogoods/nginx
[akshita@ecogoods-server ~]$ sudo chown root:srvadmin /srv
[akshita@ecogoods-server ~]$ sudo chmod 2755 /srv
[akshita@ecogoods-server ~]$ sudo setfacl -m g:sysadmins:rwx,g:srvadmin:rwx /srv
[akshita@ecogoods-server ~]$ sudo chown root:webdev /srv/http
[akshita@ecogoods-server ~]$ sudo chmod 2770 /srv/http
[akshita@ecogoods-server ~]$ sudo setfacl -m g:webdev:rwx,g:sysadmins:rwx,g:srvadmin:rwx /srv/http
[akshita@ecogoods-server ~]$ sudo setfacl -d -m g:webdev:rwx,g:sysadmins:rwx,g:srvadmin:rwx /srv/http
[akshita@ecogoods-server ~]$ echo "EcoGoods website placeholder" | sudo tee /srv/http/index.html
EcoGoods website placeholder
[akshita@ecogoods-server ~]$ sudo chown root:devteam /srv/projects
[akshita@ecogoods-server ~]$ sudo chmod 2770 /srv/projects
[akshita@ecogoods-server ~]$ sudo setfacl -m g:devteam:rwx,g:testers:rx,g:sysadmins:rwx,g:srvadmin:rwx /srv/projects
[akshita@ecogoods-server ~]$ sudo setfacl -d -m g:devteam:rwx,g:testers:rx,g:sysadmins:rwx,g:srvadmin:rwx /srv/projects
[akshita@ecogoods-server ~]$ sudo chown root:devteam /srv/projects/ecogoods
[akshita@ecogoods-server ~]$ sudo chmod 2770 /srv/projects/ecogoods
[akshita@ecogoods-server ~]$ sudo setfacl -m g:devteam:rwx,g:testers:rx,g:sysadmins:rwx,g:srvadmin:rwx /srv/projects/ecogoods
[akshita@ecogoods-server ~]$ sudo setfacl -d -m g:devteam:rwx,g:testers:rx,g:sysadmins:rwx,g:srvadmin:rwx /srv/projects/ecogoods
[akshita@ecogoods-server ~]$ echo "EcoGoods development project" | sudo tee /srv/projects/ecogoods/README.md
EcoGoods development project
[akshita@ecogoods-server ~]$ sudo chown -R root:srvadmin /srv/docker
[akshita@ecogoods-server ~]$ sudo chmod -R 2770 /srv/docker
[akshita@ecogoods-server ~]$ sudo setfacl -R -m g:sysadmins:rwx,g:srvadmin:rwx /srv/docker
[akshita@ecogoods-server ~]$ sudo setfacl -R -d -m g:sysadmins:rwx,g:srvadmin:rwx /srv/docker
[akshita@ecogoods-server ~]$
```

Figure G.1: Creation of /srv directories and application of POSIX ACLs for role-based access control

```
[akshita@ecogoods-server ~]$ ls -ld /srv /srv/http /srv/projects /srv/projects/ecogoods /srv/docker
drwxrwsr-x+ 6 root srvadmin 4096 May 10 16:27 /srv
drwxrws---+ 3 root srvadmin 4096 May 10 16:27 /srv/docker
drwxrws---+ 2 root webdev 4096 May 10 16:30 /srv/http
drwxrws---+ 3 root devteam 4096 May 10 16:26 /srv/projects
drwxrws---+ 2 root devteam 4096 May 10 16:35 /srv/projects/ecogoods
[akshita@ecogoods-server ~]$
```

Figure G.2: Directory ownership and permissions verified using ls -ld for /srv folders

```
[akshita@ecogoods-server ~]$ getfacl /srv/http
getfacl: Removing leading '/' from absolute path names
# file: srv/http
# owner: root
# group: webdev
# flags: -s-
user::rwx
group::rwx
group:sysadmins:rwx
group:srvadmin:rwx
group:webdev:rwx
mask::rwx
other::---
default:user::rwx
default:group::rwx
default:group:sysadmins:rwx
default:group:srvadmin:rwx
default:group:webdev:rwx
default:mask::rwx
default:other::---

[akshita@ecogoods-server ~]$
```

Figure G.3: ACL permissions for /srv/http verified using getfacl

```
[akshita@ecogoods-server ~]$ getfacl /srv/projects/ecogoods
getfacl: Removing leading '/' from absolute path names
# file: srv/projects/ecogoods
# owner: root
# group: devteam
# flags: -s-
user::rwx
group::rwx
group:sysadmins:rwx
group:srvadmin:rwx
group:devteam:rwx
group:testers:r-x
mask::rwx
other::---
default:user::rwx
default:group::rwx
default:group:sysadmins:rwx
default:group:srvadmin:rwx
default:group:devteam:rwx
default:group:testers:r-x
default:mask::rwx
default:other::---

[akshita@ecogoods-server ~]$
```

Figure G.4: ACL permissions for /srv/projects/ecogoods verified using getfacl

```
[akshita@ecogoods-server ~]$ getfacl /srv/docker
getfacl: Removing leading '/' from absolute path names
# file: srv/docker
# owner: root
# group: srvadmin
# flags: -s-
user::rwx
group::rwx
group:sysadmins:rwx
group:srvadmin:rwx
mask::rwx
other:---
default:user::rwx
default:group::rwx
default:group:sysadmins:rwx
default:group:srvadmin:rwx
default:mask::rwx
default:other:---

[akshita@ecogoods-server ~]$
```

Figure G.5: ACL permissions for /srv/docker verified using getfacl

```
[akshita@ecogoods-server ~]$ sudo -u avery bash -c 'echo "avery project test" > /srv/projects/ecogoods/avery.txt'
[sudo] password for akshita:
[akshita@ecogoods-server ~]$ sudo -u avery bash -c 'echo "avery web test" > /srv/http/avery.html'
[akshita@ecogoods-server ~]$ ls -l /srv/projects/ecogoods
total 8
-rw-rw----+ 1 avery devteam 19 May 10 16:47 avery.txt
-rw-rw----+ 1 root devteam 29 May 10 16:35 README.md
[akshita@ecogoods-server ~]$ ls -l /srv/http
total 8
-rw-rw----+ 1 avery webdev 15 May 10 16:48 avery.html
-rw-rw----+ 1 root webdev 29 May 10 16:30 index.html
[akshita@ecogoods-server ~]$
```

Figure G.6: Avery's full /srv access verified by writing to both project and website directories

```
[akshita@ecogoods-server ~]$ sudo -u evelynn bash -c 'echo "evelynn web test" > /srv/http/evelynn.html'
[akshita@ecogoods-server ~]$ sudo -u evelynn bash -c 'echo "should fail" > /srv/projects/ecogoods/evelynn.txt'
bash: line 1: /srv/projects/ecogoods/evelynn.txt: Permission denied
[akshita@ecogoods-server ~]$
```

Figure G.7: Evelyn's access verified by successful website write and denied project write

```
[akshita@ecogoods-server ~]$ sudo -u james cat /srv/projects/ecogoods/README.md
EcoGoods development project
[akshita@ecogoods-server ~]$ sudo -u james bash -c 'echo "should fail" > /srv/projects/ecogoods/james.txt'
bash: line 1: /srv/projects/ecogoods/james.txt: Permission denied
[akshita@ecogoods-server ~]$
```

Figure G.8: James's tester access verified by successful project read and denied project write

Appendix H - Docker Access Control, Compose Configuration and Service Verification

```
[akshita@ecogoods-server ~]$ sudo systemctl enable --now docker
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' -> '/usr/lib/systemd/system/docker.service'.
[akshita@ecogoods-server ~]$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-10 16:55:05 +0545; 14s ago
     Invocation: e80edd7b2ca54852a229f80623d050c5
   TriggeredBy: ● docker.socket
      Docs: https://docs.docker.com
    Main PID: 3042 (dockerd)
       Tasks: 9
    Memory: 108.5M (peak: 111.8M)
         CPU: 41ms
    CGroup: /system.slice/docker.service
            └─3042 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.390159432+05:45" level=info msg="Deleting nftables IPv4 rules" error="exit status 1" output="Error: C_ocker-bridges"
May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.395880547+05:45" level=info msg="Deleting nftables IPv6 rules" error="exit status 1" output="Error: C_ocker-bridges"
May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.657968486+05:45" level=info msg="Loading containers: done."
May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.666752581+05:45" level=info msg="Docker daemon" commit=56be73107b containerd-snapshotter=true storage_version=29.4.3
May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.666900876+05:45" level=info msg="Initializing buildkit"
May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.674669799+05:45" level=warning msg="git source cannot be enabled: failed to find git binary: exec: \"_ound in $PATH"
May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.677157452+05:45" level=info msg="Completed buildkit initialization"
May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.682272724+05:45" level=info msg="Daemon has completed initialization"
May 10 16:55:05 ecogoods-server dockerd[3042]: time="2026-05-10T16:55:05.682301594+05:45" level=info msg="API listen on /run/docker.sock"
May 10 16:55:05 ecogoods-server systemd[1]: Started Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
[akshita@ecogoods-server ~]$
```

Figure H.1: Docker service enabled and verified as active

```
[akshita@ecogoods-server ~]$ sudo docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

[akshita@ecogoods-server ~]$
```

Figure H.2: Docker installation tested using hello-world

```
[akshita@ecogoods-server ~]$ sudo docker ps -a
```

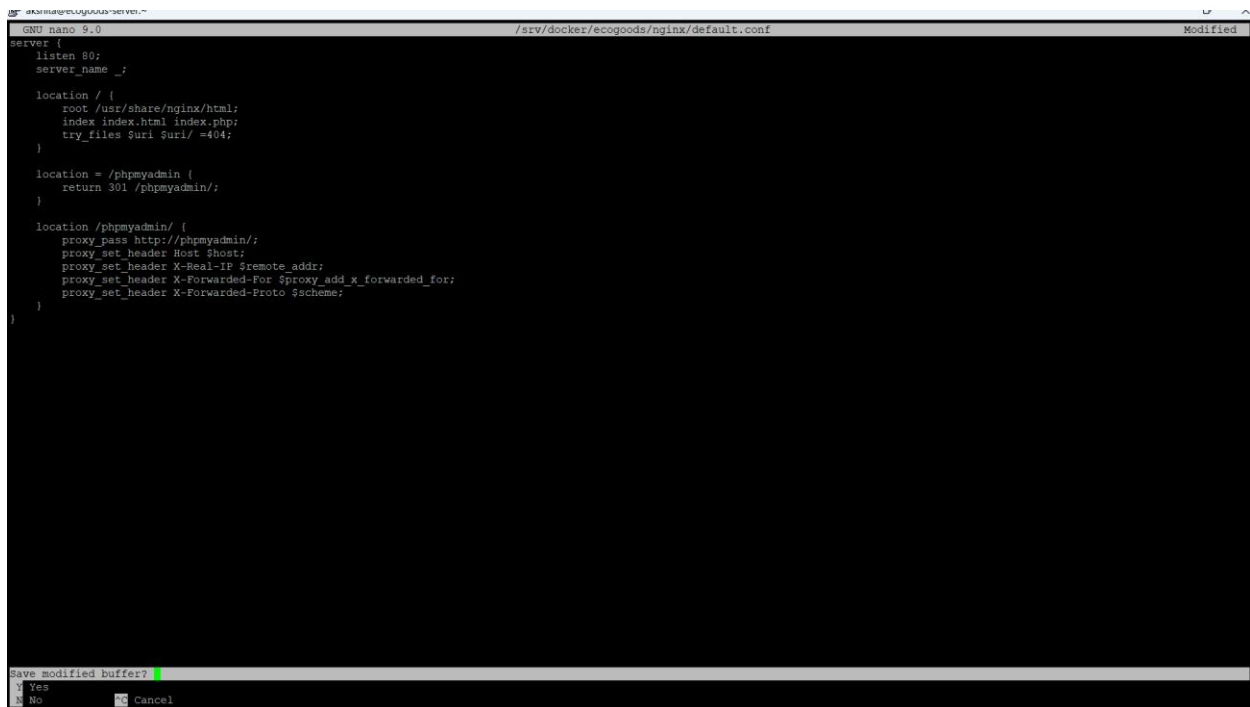
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ffd8f95c4131	hello-world	"/hello"	24 seconds ago	Exited (0) 24 seconds ago		peaceful_mestorf
cf501c6742a2	hello-world	"/hello"	About a minute ago	Exited (0) About a minute ago		gallant_bhabha

```
[akshita@ecogoods-server ~]$
```

Figure H.3: Docker container list verified using docker ps -a

```
[akshita@ecogoods-server ~]$ sudo -u evelynn docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
[akshita@ecogoods-server ~]$ sudo -u james docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
[akshita@ecogoods-server ~]$
```

Figure H.4: Docker access denied for Evelyn and James

A screenshot of a terminal window with a dark background. At the top, the title bar shows 'akshita@ecogoods-server' and the file path '/srv/docker/ecogoods/nginx/default.conf'. The terminal displays the content of the 'default.conf' file in a light-colored font. The configuration includes a 'server' block with 'listen 80;' and 'server_name _;'. It has two 'location' blocks: one for the root path '/' pointing to '/usr/share/nginx/html' with 'index index.html index.php;' and 'try_files \$uri \$uri/ =404;'; and another for '/phpmyadmin/' which uses 'proxy_pass http://phpmyadmin/;' and various 'proxy_set_header' directives. At the bottom, a prompt 'Save modified buffer?' is visible with 'Y Yes' and 'N No' options, and a 'Cancel' button.

```
GNU nano 5.0 /srv/docker/ecogoods/nginx/default.conf Modified
server {
    listen 80;
    server_name _;

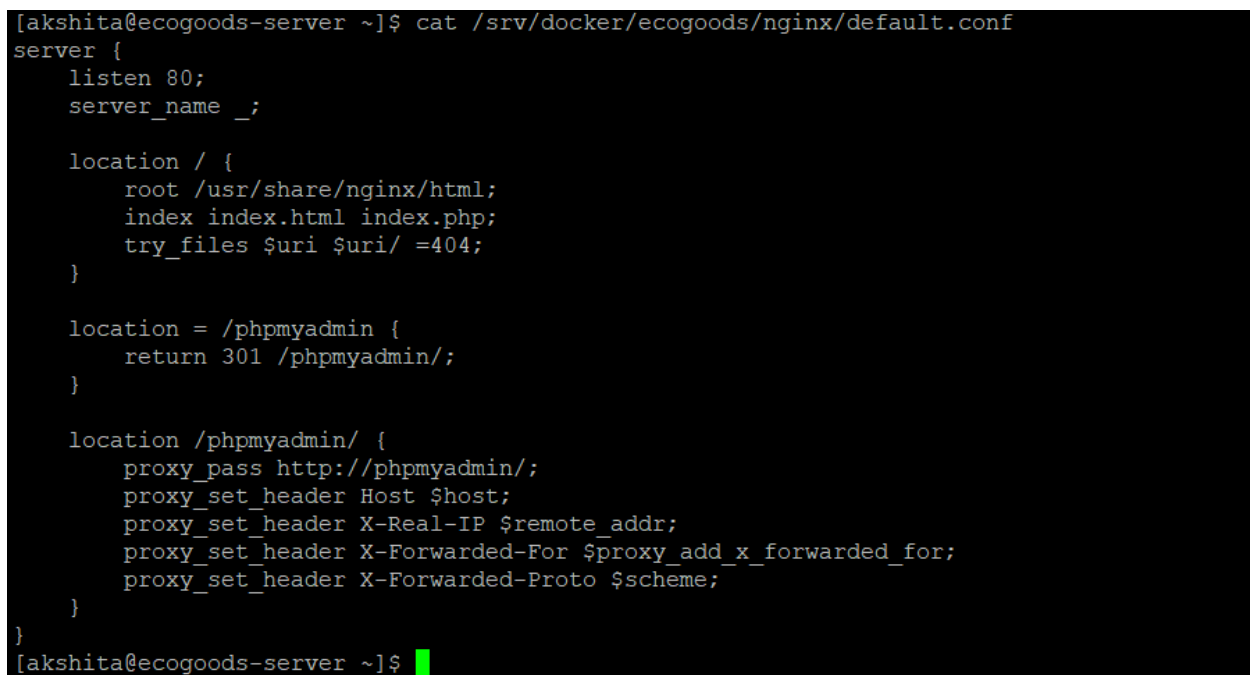
    location / {
        root /usr/share/nginx/html;
        index index.html index.php;
        try_files $uri $uri/ =404;
    }

    location = /phpmyadmin {
        return 301 /phpmyadmin/;
    }

    location /phpmyadmin/ {
        proxy_pass http://phpmyadmin/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}

Save modified buffer?
Y Yes
N No Cancel
```

Figure H.5: Nginx reverse proxy configuration created for website and /phpmyadmin/ route

A screenshot of a terminal window with a dark background. The terminal shows a command prompt '[akshita@ecogoods-server ~]\$' followed by the command 'cat /srv/docker/ecogoods/nginx/default.conf'. Below the command, the entire content of the 'default.conf' file is displayed in a light-colored font, matching the configuration shown in Figure H.5. The prompt '[akshita@ecogoods-server ~]\$' is visible at the bottom of the terminal.

```
[akshita@ecogoods-server ~]$ cat /srv/docker/ecogoods/nginx/default.conf
server {
    listen 80;
    server_name _;

    location / {
        root /usr/share/nginx/html;
        index index.html index.php;
        try_files $uri $uri/ =404;
    }

    location = /phpmyadmin {
        return 301 /phpmyadmin/;
    }

    location /phpmyadmin/ {
        proxy_pass http://phpmyadmin/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
[akshita@ecogoods-server ~]$
```

Figure H.6: Nginx reverse proxy configuration verified using cat default.conf

```
GNU nano 2.9.0 /srv/docker/ecogoods/compose.yml Modified
- internal
- web

nginx:
  image: nginx:stable-alpine
  container_name: ecogoods_nginx
  restart: unless-stopped
  ports:
    - "80:80"
  volumes:
    - /srv/http:/usr/share/nginx/html:ro
    - ./nginx/default.conf:/etc/nginx/conf.d/default.conf:ro
  depends_on:
    - phpmyadmin
  networks:
    - web

volumes:
  mysql_data:

networks:
  internal:
    internal: true
  web:
```

Figure H.7: Docker Compose file created for MySQL, phpMyAdmin and Nginx services

```

services:
  mysql:
    image: mysql:8.4
    container_name: ecogoods_mysql
    restart: unless-stopped
    environment:
      MYSQL_ROOT_PASSWORD: ChangeThisStrongRootPassword123!
      MYSQL_DATABASE: ecogoods
      MYSQL_USER: avery_dbadmin
      MYSQL_PASSWORD: ChangeThisStrongAveryPassword123!
    volumes:
      - mysql_data:/var/lib/mysql
    networks:
      - internal
    expose:
      - "3306"

  phpmyadmin:
    image: phpmyadmin:latest
    container_name: ecogoods_phpmyadmin
    restart: unless-stopped
    environment:
      PMA_HOST: mysql
      PMA_PORT: 3306
      PMA_ABSOLUTE_URI: http://localhost/phpmyadmin/
    depends_on:
      - mysql
    networks:
      - internal
      - web

  nginx:
    image: nginx:stable-alpine
    container_name: ecogoods_nginx
    restart: unless-stopped
    ports:
      - "80:80"
    volumes:
      - /srv/http:/usr/share/nginx/html:ro
      - ./nginx/default.conf:/etc/nginx/conf.d/default.conf:ro
    depends_on:
      - phpmyadmin
    networks:
      - web

volumes:
  mysql_data:

networks:
  internal:
    internal: true
  web:

```

Figure H.8: Docker Compose configuration verified using `cat compose.yml`

```
[akshita@ecogoods-server ~]$ ls -l /srv/docker/ecogoods
total 12
-rw-rw----+ 1 root srvadmin 1086 May 10 17:04 compose.yml
drwxrws---+ 2 root srvadmin 4096 May 10 17:03 nginx
[akshita@ecogoods-server ~]$ ls -l /srv/docker/ecogoods/nginx
total 4
-rw-rw----+ 1 root srvadmin 526 May 10 17:03 default.conf
[akshita@ecogoods-server ~]$
```

Figure H.9: Docker configuration file locations verified using ls -l

```
[akshita@ecogoods-server ~]$ cd /srv/docker/ecogoods
[akshita@ecogoods-server ecogoods]$ sudo docker compose up -d
[sudo] password for akshita:
[+] up 53/53
✓ Image phpmyadmin:latest      Pulled
✓ Image nginx:stable-alpine    Pulled
✓ Image mysql:8.4              Pulled
✓ Network ecogoods web         Created
✓ Network ecogoods internal     Created
✓ Volume ecogoods_mysql_data    Created
✓ Container ecogoods_mysql      Started
✓ Container ecogoods_phpmyadmin Started
✓ Container ecogoods_nginx      Started
[akshita@ecogoods-server ecogoods]$ sudo docker compose ps
[sudo] password for akshita:
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
ecogoods_mysql      mysql:8.4            "docker-entrypoint.s..." mysql      4 minutes ago  Up 4 minutes  3306/tcp, 33060/tcp
ecogoods_nginx      nginx:stable-alpine  "/docker-entrypoint..." nginx      4 minutes ago  Up 4 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp
ecogoods_phpmyadmin phpmyadmin:latest    "/docker-entrypoint..." phpmyadmin 4 minutes ago  Up 4 minutes  80/tcp
[akshita@ecogoods-server ecogoods]$ sudo docker ps
CONTAINER ID        IMAGE                COMMAND                CREATED        STATUS        PORTS                NAMES
5bc087c24406        nginx:stable-alpine  "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp  ecogoods_nginx
ddbdcba08c4e        phpmyadmin:latest    "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  80/tcp                ecogoods_phpmyadmin
e7f771649fd9        mysql:8.4            "docker-entrypoint.s..." 6 minutes ago  Up 6 minutes  3306/tcp, 33060/tcp    ecogoods_mysql
[akshita@ecogoods-server ecogoods]$ sudo docker network ls
NETWORK ID          NAME                DRIVER    SCOPE
c37fd08a9c1a        bridge             bridge    local
a0382268f74c        ecogoods_internal  bridge    local
8e76f00f1ec0        ecogoods_web       bridge    local
da918aaeed9c        host               host      local
91e10e6d083f        none               null      local
[akshita@ecogoods-server ecogoods]$
```

Figure H.10: Docker Compose stack started and containers verified as running

```
[akshita@ecogoods-server ecogoods]$ curl http://localhost/
EcoGoods website placeholder
[akshita@ecogoods-server ecogoods]$
```

Figure H.11: Nginx website service verified using curl http://localhost/

```
[akshita@ecogoods-server ecogoods]$ curl -I http://localhost/phpmyadmin/
HTTP/1.1 200 OK
Server: nginx/1.30.0
Date: Sun, 10 May 2026 11:36:20 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
X-Powered-By: PHP/8.3.31
Set-Cookie: phpMyAdmin=a974445a0f314ea0ba92fa6e65aaec4d; path=/phpmyadmin/; HttpOnly; SameSite=Strict
Expires: Sun, 10 May 2026 11:36:20 +0000
Cache-Control: no-store, no-cache, must-revalidate, pre-check=0, post-check=0, max-age=0
Last-Modified: Sun, 10 May 2026 11:36:20 +0000
Set-Cookie: phpMyAdmin=a974445a0f314ea0ba92fa6e65aaec4d; path=/phpmyadmin/; HttpOnly; SameSite=Strict
Set-Cookie: pma_lang=en; expires=Tue, 09 Jun 2026 11:36:20 GMT; Max-Age=2592000; path=/phpmyadmin/; HttpOnly; SameSite=Strict
Set-Cookie: phpMyAdmin=74a0e422644ccadfbcb4d2b5518305ef; path=/phpmyadmin/; HttpOnly; SameSite=Strict
X-ob_mode: 1
X-Frame-Options: DENY
Referer-Policy: same-origin
Content-Security-Policy: default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsafe-eval' ;style-src 'self' 'unsafe-inline' ;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';
X-Content-Security-Policy: default-src 'self' ;options inline-script eval-script;referrer no-referrer;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';
X-WebKit-CSP: default-src 'self' ;script-src 'self' 'unsafe-inline' 'unsafe-eval';referrer no-referrer;style-src 'self' 'unsafe-inline' ;img-src 'self' data: *.tile.openstreetmap.org;object-src 'none';
X-XSS-Protection: 1; mode=block
X-Content-Type-Options: nosniff
X-Permitted-Cross-Domain-Policies: none
X-Robots-Tag: noindex, nofollow
Pragma: no-cache
Vary: Accept-Encoding

[akshita@ecogoods-server ecogoods]$
```

Figure H.12: phpMyAdmin reverse proxy route verified using `curl -I http://localhost/phpmyadmin/`

```
[akshita@ecogoods-server ecogoods]$ sudo docker port ecogoods_mysql
[akshita@ecogoods-server ecogoods]$ sudo docker inspect ecogoods_mysql --format '{{json .NetworkSettings.Ports}}'
{"3306/tcp":null,"33060/tcp":null}
[akshita@ecogoods-server ecogoods]$ ss -tulpn | grep 3306
[akshita@ecogoods-server ecogoods]$
```

Figure H.13: MySQL confirmed as not externally accessible using Docker port and host listener checks

```

[avery@ecogoods-server ~]$ docker exec -it ecogoods_mysql mysql -uavery_dbadmin
-p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.4.9 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| ecogoods |
| information_schema |
| performance_schema |
+-----+
3 rows in set (0.00 sec)

mysql> █

```

Figure H.14: Avery's internal MySQL database user verified using SHOW DATABASES

```

[akshita@ecogoods-server ecogoods]$ ss -tulpn | grep ':80'
tcp    LISTEN 0      4096      0.0.0.0:80      0.0.0.0:*
tcp    LISTEN 0      4096      [::]:80        [::]:*
[akshita@ecogoods-server ecogoods]$ █

```

Figure H.15: Nginx confirmed as the exposed web entry point on port 80

```

[akshita@ecogoods-server ecogoods]$ sudo ss -tulpn
[sudo] password for akshita:
Netid    State      Recv-Q    Send-Q          Local Address:Port    Peer Address:Port    Process
tcp      LISTEN     0          128      127.0.0.1:46813      0.0.0.0:*            users:(("containerd",pid=3022,fd=15))
tcp      LISTEN     0          128      0.0.0.0:22          0.0.0.0:*            users:(("sshd",pid=1746,fd=6))
tcp      LISTEN     0          4096      0.0.0.0:80          0.0.0.0:*            users:(("docker-proxy",pid=4041,fd=7))
tcp      LISTEN     0          128      [::]:22             [::]:*               users:(("sshd",pid=1746,fd=7))
tcp      LISTEN     0          4096      [::]:80             [::]:*               users:(("docker-proxy",pid=4049,fd=7))
[akshita@ecogoods-server ecogoods]$ █

```

Figure H.16: Host listening ports verified using sudo ss -tulpn